

Data Mining

Complete Exam Notes

CT 72502 (Elective I)

BEI / BEX · Year IV Part I

Institute of Engineering, Tribhuvan University

Built from the syllabus, the class notes, and **all 17 past papers (2070–2081 BS)**. Every topic carries a frequency tier computed from those papers, and every formula is checked against Han & Kamber, *Data Mining: Concepts and Techniques*, 3rd edition.

Numerical solutions live in the **per-chapter numerical companions**; each chapter marks its numerals with **N**.

Short on time? Go straight to §8, Last-Minute Revision.

Ch	Topic	Hours	Papers
1	Introduction	2	16/17
2	Data Preprocessing	6	17/17
3	Classification	12	17/17
4	Association Analysis	10	17/17
5	Cluster Analysis	9	17/17
6	Anomaly / Fraud Detection	3	17/17
7	Advanced Applications	3	16/17
Total		45	

Contents

1	Introduction	3
2	Data Preprocessing	12
3	Classification	24
4	Association Analysis	36
5	Cluster Analysis	43
6	Anomaly / Fraud Detection	49
7	Advanced Applications	55
8	Last-Minute Revision	59

1. Introduction

2 hrs · in 22/23 papers · 2 topics, both TOP tier · no numericals in this ch.

Whole chapter in 2 items:

- KDD process **TOP 9/23**
- DW fundamentals **TOP 12/23**

1.1 Data Mining Origin

What is DM? + Explain steps of KDD process **2+3 2+5 2+6 1+5 4**

TOP 9/23 82 Bh, 79 Ba Most repeated Q in the paper.

DM

- Extraction of **interesting** patterns/knowledge from large data.
- Interesting = non-trivial + implicit + previously unknown + potentially useful.
- Not retrieval:
 - Query tool → answers what you already know to ask.
 - DM → finds patterns you didn't know to look for.
- Also called: KDD, knowledge extraction, data/pattern analysis, data archaeology, data dredging, BI.

Not DM

- Phone-book lookup, `SELECT AVG(salary)`, web search → info retrieval (query states exactly what is wanted).

Descriptive DM

- Characterise the general properties of data.
- Unsupervised.
- “What is in data? Unusual patterns?”
- Clustering, association, summarisation, visualisation.

Predictive DM

- Infer unknown $Y = f(X)$ from known X .
- Supervised.
- “Which customers leave in 6 months?”
- Classification, regression, prediction.

DM vs KDD

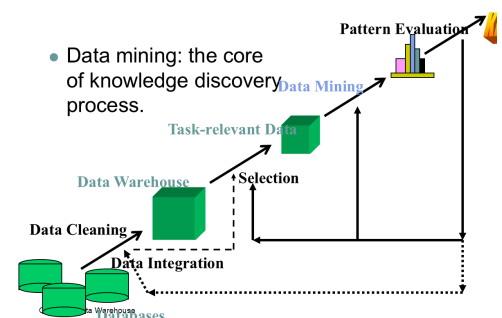
- Strictly: DM = **one step** of KDD (the step that runs the algorithms).
- In practice treated as synonyms. **IOE papers use them interchangeably.**
- If Q says “steps of data mining” → write KDD steps.

KDD: 7 steps (Han & Kamber) → default answer

1. **Data cleaning:** rm noise, smooth outliers, fill missing, fix inconsistency.
2. **Data integration:** combine multiple heterogeneous sources.
3. **Data selection:** retrieve only task-relevant data.
4. **Data transformation:** normalise, aggregate, generalise, discretise.
5. **Data mining:** apply algorithms. *Essential step.*
6. **Pattern evaluation:** keep truly interesting patterns (support, confidence, lift, novelty).
7. **Knowledge presentation:** visualise + represent for user.

Steps 1-4 = data preparation ≈ 60% of effort.

Data Mining: A KDD Process



Conflict in your sources **verified/added**

- Chapter 2_Data Warehouse.ppt → **6 stages** (Adriaans & Zantinge): selection → cleaning → enrichment → coding → mining → reporting.
- Han & Kamber (syllabus ref. book 2) → **7 steps** above.

- Both examined at IOE (old CT 725 paper asked “Cleaning, Enrichment, Coding”).
- **Write 7-step version, add 1 line naming the 6-stage variant.**

Answer skeleton [5–7]

1. DM definition, 1 line + “one step of KDD”.
2. **Draw the pipeline** (markers reward the fig).
3. 7 steps, **1 line each**. No more.
4. Close: “steps 1–4 = data prep $\approx$ 60% effort”.

Origin of DM: why it emerged 2 3

Confluence of disciplines

- DB systems
- Statistics
- Machine learning
- AI / pattern recognition
- Visualisation
- High-performance computing

Drivers

- Automated collection → data explosion (POS scanners, txn logs, sensors, satellites, Web).
- Cheap storage → hoarding became rational. Analysis did not keep pace.
- Competition → unmined data = wasted asset.

Why traditional/manual analysis fails

- Enormity (TB scale).
- **High dimensionality** (1000s of attrs).
- Sources heterogeneous + distributed.
- Data collected for **ops**, not analysis → noisy, incomplete, inconsistent.

DM vs query tools (SQL)

- SQL: must know *exactly* what you want. DM: only *vaguely*.
- SQL → facts/aggregates. DM → patterns/trends.
- SQL: deterministic retrieval. DM: ML/statistical search.
- Hidden relations:
 - SQL cant trace them
 - DM finds them automatically

“The world is data rich but information poor”. Justify. 8

PIN 1/23 73 Shr Justify-in-own-words → needs structure, not definitions.

1. **State paradox:** TBs stored, decisions still on intuition. Data $\neq$ info $\neq$ knowledge.
2. **Data rich:** POS logs, bank txns, telecom call records, remote sensing, Web. Growth automated.
3. **Info poor, why:**
 - Volume defeats manual inspection.
 - Data is operational → noisy, incomplete, inconsistent.
 - Scattered across heterogeneous systems.
 - Query tools answer only what is asked.
 - Shortage of analysts.
4. **Consequence:** “data tombs” = archives written, never read. Decisions made w/o evidence already paid for.
5. **Resolution:**
 - DW consolidates + cleans
 - DM extracts the patterns
 - Data → Info → Knowledge → Decision
6. **Eg:** retail chain holds millions of receipts (rich) but cant say which products sell together (poor) until association analysis gives {nappies} $\Rightarrow$ {beer}.

“Data mining turns a large collection of data into knowledge”. Justify 6

81 Bh The mirror image of the question above, and it is asked separately, so learn it as its own answer. That one asks why we are stuck; this one asks how DM gets us out. Structure the answer on the four-rung ladder and carry one example all the way up it. verified/added

The ladder, one rung at a time

1. **Data:** raw recorded facts, no context. Eg 2 million supermarket receipt lines.
2. **Information:** data aggregated and given context by querying. Eg “nappies sold 4,200 units last month”. A DW and OLAP get you here, and **this is as far as a query tool can go** — it answers only what you thought to ask.
3. **Knowledge:** patterns that hold across the data and **were not asked for**. Eg “customers buying nappies on Friday evening also buy beer”. Only **data mining** produces this rung, because only it searches the space of possible patterns.
4. **Decision / wisdom:** acting on the pattern. Eg move the beer display beside the nappies, and measure the lift in basket value.

Why the step to knowledge is data mining’s alone

- **Queries are deductive, mining is inductive.** SQL retrieves what you specify; mining induces a general rule from the instances.
- It is **automatic and exhaustive**: Apriori tests every itemset the thresholds allow, far more hypotheses than an analyst would think to write.
- The output is **predictive**, not just descriptive: a classifier built on past data labels tuples that do not exist yet.
- The patterns are required to be **valid, novel, useful and understandable** — Fayyad’s four conditions. **“Novel” is what makes it knowledge**: a pattern you already knew is information, not knowledge.

Close with the qualifier

- DM does not turn *all* data into knowledge. Garbage in, garbage out: without preprocessing (Ch 2) it turns noise into confident nonsense, and without pattern evaluation (§4.1) it reports rules that are strong but false.

Architecture of a typical DM system (+Fig) 2 4+2

78 Bh asked as “Explain DM process with diagram”.

- **Data sources:** DB, DW, WWW, other repos.
- **DB/DW server:** fetches data relevant to the request.
- **Knowledge base:** domain knowledge guiding search + judging interestingness (concept hierarchies, user beliefs, constraints).
- **DM engine:** characterisation, association, classification, clustering, outlier, evolution analysis.
- **Pattern evaluation:** applies interestingness measures. *Best pushed into the engine* so dull patterns are never generated.
- **UI:** specify task, browse schemas/hierarchies, visualise results.

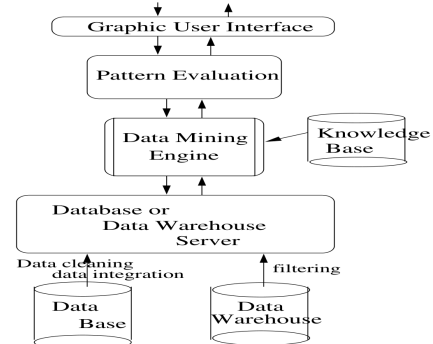


Figure 1.5: Architecture of a typical data mining system.

DM functionalities 2 4

- **Characterisation & discrimination:**
 - summarise the target class
 - contrast it against a comparative class
- **Association & correlation:** freq patterns, rules $X \Rightarrow Y$. → Ch 4
- **Classification & prediction:** assign class labels / predict numeric. → Ch 3
- **Cluster analysis:** group w/o pre-defined labels. → Ch 5
- **Outlier / anomaly:** objects not complying w/ general behaviour. Noise to discard, but *the prize* in fraud. → Ch 6
- **Evolution analysis:** trends for objects changing over time. → Ch 7

What is DM? Explain its applications. 10

PIN 1/23 71 Ch

Domain	What is mined	
Biomedical / DNA	Similarity search across DNA seqs; association analysis for co-occurring genes (most diseases = gene combinations, not 1 gene); path analysis linking genes to disease stages.	
Financial	DW design for multidim analysis; loan payment prediction, credit policy; classification + clustering for customer targeting; money-laundering detection .	
Retail / market basket	Which products sell together; customer retention; campaign effectiveness; recommendation.	
Telecom	Fraudulent usage patterns; multidim analysis of calls; mobile user behaviour.	
Intrusion detection	Anomaly detection over traffic + audit logs.	→ Ch 6
Web mining	PageRank, usage + structure mining.	→ Ch 7

1.2 Data Mining & Data Warehousing Basics

What is a DW? Write its key features. 2 2+4 2+5

TOP 12/23 73 Ch, 70 Asa, 69 Ch

Definition (Inmon)

- DW = **subject-oriented, integrated, time-variant, non-volatile** collection of data supporting management decision-making.
- **Memorise as S-I-T-N** → this is the whole “key features” answer.
- **S = Subject-oriented**
 - Organised around subjects (customer, product, sales), not daily ops.
 - Data useless for decisions is excluded.
- **I = Integrated**
 - Built from heterogeneous sources (relational DBs, flat files, txn records).
 - Cleaning + integration force consistent naming, encoding, units.
 - → 1 name per entity.
- **T = Time-variant**
 - Long history (5–10 yrs).
 - Every key structure carries a time element.
 - Operational DB holds only the *current* value.
- **N = Non-volatile**
 - Physically separate. No operational update.
 - Only 2 ops: **initial load** + **access**.
 - → no txn processing, recovery or concurrency control needed.
- **Data warehousing** = the *process* of building + using a DW (extract, transform operational data, load into central store).

How is DW diff from a DB? How are they similar? 2+2

PIN 3/23 75 Ash, 73 Ch Asks both directions. 73 Ch asks it as DW vs RDBMS.

	Operational DB (OLTP)	Data warehouse (OLAP)
Purpose	Run day-to-day business	Support analysis + decisions
Content	Current, detailed	Historical, summarised + detailed
Design	Highly normalised, many tables (ER)	De-normalised, few tables; star/snowflake
Access	Short read/write txns	Long, complex, read-mostly
Users	Clerks, DBAs, front-line	Managers, analysts, execs
Unit of work	Simple txn	Complex query
Records/query	Tens	Millions
Size	100 MB–GB	100 GB–TB

Similarities verified/added

- Both persistent, managed data stores.
- Both usually on a relational DBMS.
- Both use schemas, indexes, SQL.
- Both enforce integrity + security.
- DW is *built from* the DBs.

Why keep them separate

- **Performance:** opposite tuning.
 - DBMS → OLTP (indexing, concurrency, recovery)
 - DW → OLAP (multidim views, consolidation)
 - both on 1 system degrades both
- **Missing data:** decisions need history ops DBs dont keep.
- **Consolidation:** needs aggregation across heterogeneous sources.
- **Data quality:** inconsistent codes/formats reconciled once, centrally.

Explain DW architecture with its analytical processing 8

TOP 9/23 75 Ch 2 architectures in your sources: complementary, not contradictory (three-tier = layered, process = component-wise).

(a) Three-tier architecture (Han & Kamber) → standard answer

- **Bottom tier** = warehouse DB server (relational).
 - Data pulled via **gateways**: ODBC, OLE-DB, JDBC.
 - Back-end tools: extract, clean, transform, load, refresh.
 - Holds **metadata repo**.
- **Middle tier** = **OLAP server** (presents data as a cube).
 - **ROLAP:**
 - o extended relational DBMS
 - o maps multidim ops → relational ops
 - o scales to large data
 - **MOLAP:**
 - o special-purpose server, direct multidim arrays
 - o fast, but sparse data wastes space
 - **HOLAP:**
 - o hybrid of the two
 - o detail in relational, aggregates in arrays
- **Top tier** = front-end client: query & reporting, analysis, DM tools.

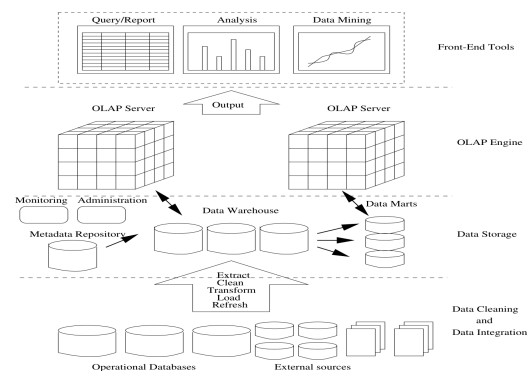
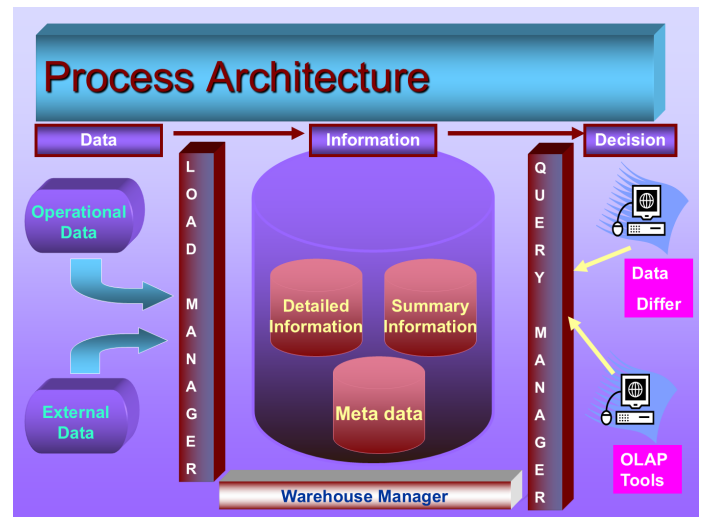


Figure 2.12: A three-tier data warehousing architecture.

(b) Process / component architecture (form drawn in class notes)

- **Load manager:** all extract + load ops. Fast-loads into temp store, simple transforms.
- **Warehouse manager:**
 - consistency + referential checks
 - merges temp store into the published DW
 - creates indexes + business views
 - updates aggregations
 - backup + archive
- **Query manager:**
 - directs queries to the right tables
 - stops 1 query starving the system
 - captures query profiles
- **Detailed info:** lowest-granularity data retained.
- **Summary info:**
 - pre-computed aggregations built by the warehouse manager
 - transient, changes as query profiles change
- **Metadata:** data about data. Describes how info is structured, maps sources to a common view.



(c) "...with its analytical processing" → 3 kinds of DW application

1. **Information processing:** querying, basic stats, reporting (crosstabs, tables, charts). *No new knowledge.*
2. **Analytical processing (OLAP):** multidim analysis. Roll-up, drill-down, slice, dice, pivot. *Summarises, doesn't learn.*
3. **Data mining:** knowledge discovery from hidden patterns. *Creates new knowledge.*

What is DW and data mart? (short notes) 15 2+4

PIN 3/23 70 Ch, 71 Shr, 69 Ch

Data mart

- Subset of a DW, confined to 1 user group / functional area, in its own DB.
- **Dependent:** sourced *from* an enterprise DW → consistent by construction.
- **Independent:** populated *directly* from sources, no central DW → faster to build, risks inconsistency between marts.
- **Benefit:**
 - less volume to scan → faster queries
 - each department owns its own structures
- **Eg:** departmental store sales mart, bank loan mart.

3 DW models

- **Enterprise warehouse:** all subjects, whole org. Cross-functional, large, slow to build.
- **Data mart:** subset for a specific group.
- **Virtual warehouse:**
 - set of views over operational DBs
 - only some summary views materialised
 - easy to build, but loads the operational systems

Describe Snowflake schema with example 4

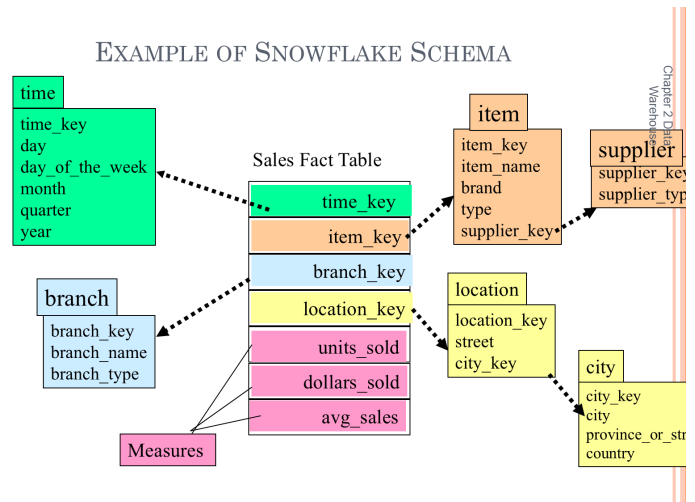
PIN 1/23 74 Ch

- **Star**: 1 central **fact table** (measures + FKs) + **de-normalised** dimension tables. Simple, fast joins.
- **Snowflake**: star refined, some dimension hierarchies **normalised** into smaller sub-dimension tables. Saves space, less redundancy, but more joins → slower.
- **Fact constellation (galaxy)**: multiple fact tables sharing dimension tables = collection of stars.

Eg: sales DW

- Fact Sales: time_key, item_key, branch_key, location_key, units_sold, dollars_sold, avg_sales.
- location normalised: location → city → province.
- item → supplier.
- **That normalisation of the dimension hierarchy is what makes it snowflake, not star.**

→ Full multidim model (cuboids, base/apex, star-schema drawing numerical) sits in **Ch 2, OLAP & Multidimensional Data Analysis**, where it is examined harder.



Fundamental differences between DM and DW 3

PIN 2/23 76 Ash

	Data warehousing	Data mining
Nature	Storage architecture	Analysis process
Purpose	Consolidate + organise data	Discover hidden patterns in it
Output	Consistent query-able historical data	Rules, models, clusters, predictions
Done by	Engineers / DBAs	Analysts, statisticians, business users
Relation	DW = infrastructure; DM = 1 of its 3 uses. DW not required for DM, but a clean DW makes DM far more effective.	

How can a DM system be integrated with a DB/DW system? 4

PIN 1/23 78 Bh Answer = the 4 coupling schemes.

1. No coupling

- Uses no DB/DW function at all.
- Reads flat file → mines → writes to another file.
- Simple, but: poor data selection, no scalability, no consistency.

2. Loose coupling

- Mining system → fetches data from repo managed by DB/DW.
- Stores results back into a file / designated place in DB.
- Uses DBMS for storage + retrieval only.
- Cant exploit indexing or query optimisation → memory-bound, doesnt scale well.

3. Semi-tight coupling

- Besides linking, DB/DW provides efficient **mining primitives**.
- Primitives: sorting, indexing, aggregation, histogram analysis, multi-way join.
- + pre-computed stats: sum, count, max, min.
- → materially better performance.

4. Tight coupling

- DM system **smoothly integrated** into DB/DW as 1 functional component.
- Mining queries optimised alongside DB queries.
- Intermediate results can be stored.
- Scales. **Most desirable design.**

1.3 PYQ Mapping

Tiers: **TOP /23** = 9+/23 papers **HOT /23** = 5-8 **PIN /23** = 1-4 but ≥ 8 marks. Same Q in diff wording grouped under 1 entry.

1.1 Data Mining Origin

- TOP 7/23** Define DM + explain KDD / DM process. 2+3 to 2+6
 - "What is data mining? Explain the process of data mining." 74 Ash
 - "...Explain all the steps of knowledge discovery." 72 Ch
 - "...What are the steps involved in knowledge discovery process?" 79 Bh
 - "...Explain the steps of KDD process briefly." 81 Ba
 - "...Explain general steps in brief." 72 Ka
 - "What is a Data Mining? Explain its application." 71 Ch → 2nd half swaps to applications.
- PIN 1/23** "The world is data rich but information poor". Justify. 8 73 Shr

1.2 DM & DW basics

- HOT 3/23** What is a DW / key features + architecture.
 - "What is data warehousing? Explain with example where data warehouses are used." 2+4 80 Ba
 - "Write key features of data warehouse. Explain each step of the KDD process w/ example." 2+5 80 Bh
 - "Explain Data Warehouse architecture with its analytical processing." 8 75 Ch
- PIN 2/23** DM vs DW / DW vs DB.
 - "What are the fundamental differences between Data Mining and Data Warehousing? Describe the steps of KDD." 3+7 76 Ash
 - "How is data warehouse different from a database? How are they similar?" 2+2 75 Ash
- PIN 1/23** "Explain how DM system can be integrated with DB/DW system. Explain DM process with diagram." 4+2 78 Bh
- PIN 1/23** "What is data warehouse and data mart? Describe Snowflake scheme with example." 2+4 74 Ch
- PIN 2/23** Short notes: Data warehouse and Data mart. 15 70 Ch, 71 Shr

Papers added by the 2026-09 merge

The six papers of DM4.1BCT.pdf (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. verified/added

- What is data mining? Describe the steps involved in it. 2+6 82 Bh
- "Data mining turns a large collection of data into knowledge". Justify. 6 81 Bh verified/added
- KDD process + **the challenges that motivated data mining.** 3+3 79 Ba
- How is a data warehouse different from an RDBMS? Also list the similarities. 2+2 73 Ch
- Short notes: **Data warehouse** 70 Asa · **Datamart** 69 Ch. 5 5

Coverage check

- All Ch 1 PYQs covered above.
- Filled by research, absent from your notes verified/added :
 - Similarity half of "how are DW and DB similar?" 75 Ash (notes only contrast them).
 - Reconciliation of the **6-stage vs 7-step KDD** clash between your 2 sources.

Sources used for this chapter

- **Han & Kamber 3rd ed** (full textbook PDF): ch 1 KDD process, DM system architecture, functionalities, applications; ch 4 DW definition, 3-tier architecture, DW models, DW usage. **Primary authority; all of the above re-verified against it.**
- **Han & Kamber 1st ed manuscript** (ch01.pdf, ch2.pdf): source of the two architecture figures, and of the **4 coupling schemes**, which the 3rd ed drops. **78 Bh still examines coupling**, so it is kept here.
- Chapter 2_Data Warehouse.ppt, Chapter 4 and 5 DW Architecture.ppt, applicationDM.ppt, data mining_/Chapter 1_2.pdf.
- **No DBK lecture exists for Unit-1**, so class slides could not be used here.

Numericals

- **Ch 1 has no numerical component.** No Q in 23 papers needs a calculation here → no “Ch1 Numerical” companion. First numericals appear in Ch 2 (normalisation, similarity measures, PCA).

2. Data Preprocessing

6 hrs · in 23/23 papers · 4 topics, 2 TOP tier · first chapter with numericals → see the Ch 2 Numerical companion.

Weight of the chapter:

- Data Pre-processing **TOP 12/23** → most-asked topic in the whole syllabus.
- OLAP & Multidim Analysis **TOP 10/23**
- Similarity Measures **HOT 6/23** → nearly always a numerical.
- Data Types & Attributes **PIN 1/23**

2.1 Data Types and Attributes

What is DM? Explain diff data types of attributes in a dataset. **10**

PIN 1/23 71 Shr DM definition → Ch 1. Only attribute types needed here.

Attribute

- Property / characteristic of an object. Eg eye colour, temperature.
- Also called: variable, field, feature, characteristic.
- **Object** = record, point, case, sample, entity, instance.
- Same attr → diff value sets (height in ft or m).
- Diff attrs → same value set, diff properties (ID and age are both int, but ID has no max).

Han & Kamber 3rd ed: 4 types (use this, it is the syllabus ref book)

- **Nominal**: names / categories only, no order. Zip code, employee ID, eye colour.
- **Binary**: nominal w/ exactly 2 states (0/1).
 - **Symmetric**: both states equally valuable. Gender.
 - **Asymmetric**: one state rarer + more important. Disease test P/N. **Convention: rarest outcome = 1.**
- **Ordinal**: order meaningful, gap between successive values unknown. {small, medium, large}, grades, mineral hardness.
- **Numeric**: measurable quantity.
 - **Interval-scaled**: gaps meaningful, **no true zero**. Dates, temp in °C/°F.
 - **Ratio-scaled**: gaps and ratios meaningful, true zero. Temp in K, money, counts, age.

Older N-O-I-R split (Tan; DBK's slide)

- Nominal, Ordinal, Interval, Ratio. Binary folded into nominal.
- **Same content, different grouping**. Either scores; H&K's is safer as it names binary explicitly, which the similarity questions depend on.

Attribute Level	Transformation	Comments
Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
Ordinal	An order preserving change of values, i.e., $new_value = f(old_value)$ where f is a monotonic function.	An attribute encompassing the notion of good, better, best can be represented equally well by the values {1, 2, 3} or by {0.5, 1, 10}.
Interval	$new_value = a * old_value + b$ where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
Ratio	$new_value = a * old_value$	Length can be measured in meters or feet.

Prepared by: Er. Dinesh Baniya Kshatri

2nd cut: discrete vs continuous

- **Discrete:** finite / countably infinite set. Zip codes, counts, words in a doc. Usually int.
 - **Binary** = special case of discrete.
 - **Symmetric binary:** both states equally valuable (gender).
 - **Asymmetric binary:** one state rarer + more important (disease test P/N). **Matters: Jaccard ignores 0-0 matches, SMC does not.**
- **Continuous:** real values. Temp, height, weight. Stored as float.

Types of datasets

- **Record**
 - *Data matrix:* m objects $\times$ n numeric attrs $\rightarrow$ points in n -dim space.
 - *Document data:* each doc = term vector, value = term frequency.
 - *Transaction data:* each record = set of items. Eg grocery basket.
- **Graph:** nodes + links. Eg WWW, molecular structures.
- **Ordered:** spatial (geo coords), temporal ($d = f(t)$), sequential, sequence data.

Characteristics of structured data

- **Dimensionality** $\rightarrow$ curse of dimensionality (see 2.2).
- **Sparsity:** % of cells not populated. Sparsity + density = 100.
- **Resolution:** patterns depend on the scale you look at.

2.2 Data Pre-processing

What is Data Pre-Processing? Explain the major tasks performed in it. 2+7 4+4 5+5

TOP 16/23 82 Bh, 81 Ba, 79 Ba, 75 Ch, 73 Ch, 72 Ka, 69 Ch Most-asked topic in the syllabus.

Why preprocess

- Real data is **dirty**: incomplete, noisy, inconsistent.
- Cause: huge size + multiple heterogeneous sources.
- **Garbage in $\rightarrow$ garbage out.** Low quality data $\rightarrow$ low quality mining.
- Data prep $\approx$ **60% of total effort** in KDD.

Measures of data quality 73 Shr asks this

- **Accuracy:** values correct.
- **Completeness:** no missing values.
- **Consistency:** no contradictions.
- **Timeliness:** updated in time.
- **Believability:** how far users trust it.
- **Interpretability:** how easily understood.

5 major tasks (this is the answer)

1. **Data cleaning:** fill missing, smooth noise, fix inconsistency, rm duplicates.
2. **Data integration:** merge multiple sources into 1 coherent store.
3. **Data transformation:** normalise, aggregate, generalise, construct attrs.
4. **Data reduction:** smaller volume, same analytical result.
5. **Data discretisation:** continuous $\rightarrow$ intervals (part of reduction).

Answer skeleton 7

1. Definition + "why": dirty data, garbage in garbage out.
2. Name the 5 tasks.
3. 2 lines each, with 1 concrete technique per task.
4. Close: cleaning + integration dominate the effort.

(a) Data cleaning

Missing values: why

- Equipment malfunction, misunderstanding at entry, deleted as inconsistent, not considered important, no change recorded.

How to handle missing data 74 Ash, 74 Ch

1. **Ignore the tuple:** usual when class label missing. Bad if % missing varies a lot.
2. **Fill manually:** tedious, infeasible at scale.
3. **Global constant:** eg "Unknown". Risk: mining may treat "Unknown" as a real class.
4. **Attribute mean** of all samples.
5. **Attribute mean of same class** → better than plain mean.
6. **Most probable value:** regression, Bayesian inference, decision tree. **Most popular + most accurate.**

Noisy data

- Noise = random error / variance in a measured variable.
- Cause: faulty instruments, entry error, transmission error, tech limits, naming inconsistency.

How to handle noise

- **Binning:** sort, partition into bins, smooth by bin *mean* / *median* / *boundary*.
- **Regression:** fit data to a function.
- **Clustering:** detect + rm outliers.
- **Combined computer + human:** machine flags suspicious, human checks.

Binning: 2 partitioning rules

- **Equal-width (distance):** N intervals of equal size.
- **Equal-depth (frequency):** N intervals, each with $\approx$ same no. of samples. Better for skewed data.

Equal-width bin width, A = lowest value, B = highest, N = no. of bins:

$$W = \frac{B - A}{N}$$

Your class notes print $W = (A - B)/N$, which gives a negative width. Wrong: it is $(B - A)/N$. verified/added

(b) Data integration 5

75 Ash

- Combines data from multiple sources into 1 coherent store.
- **Issues:**
 - **Entity identification:** is `cust_id` in A the same as `cust_no` in B? Solved w/ metadata.
 - **Redundancy:** an attr derivable from another. Detect by **correlation analysis** (χ^2 for nominal, correlation coefficient + covariance for numeric).
 - **Data value conflicts:** same entity, diff values across sources (diff units, scales, currency, representation).
 - **Duplication** at tuple level.
- Careful integration → reduces/avoids redundancy → improves mining speed + quality.

χ^2 test of independence (nominal pairs) NEW

- H_0 : the 2 attrs are **independent**. H_1 : associated (so 1 is redundant).
- Build the **contingency table** of observed counts O_{ij} .
- Expected count $E_{ij} = \frac{(\text{row}_i \text{ total}) \times (\text{col}_j \text{ total})}{\text{grand total}}$
- $\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$, $df = (r - 1)(c - 1)$.
- $\chi^2 >$ critical value at $\alpha \rightarrow$ **reject** $H_0 \rightarrow$ correlated $\rightarrow$ drop one attr.

Worked in one line

- Gender (M/F) vs preference (like/dislike), 100 people, all 4 margins = 50.
- Every $E_{ij} = \frac{50 \times 50}{100} = 25$. Observed 20/30/30/20.
- $\chi^2 = 4 \times \frac{5^2}{25} = 4$, $df = (2 - 1)(2 - 1) = 1$.
- Critical value at $\alpha = 0.05$, $df = 1$ is 3.841. $4 > 3.841 \rightarrow$ **reject** H_0 , gender and preference are associated.
- **Correlation $\neq$ causation.** Numeric analogue:
 $r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B}$, formula in §2.4.

(c) Data transformation + normalisation 2+6 4 2+2+2

TOP /23 79 Ba, 78 Bh, 75 Ch, 73 Shr Normalisation numerical is a recurring 6-mark question → full solution in the Numerical companion.

Transformation methods

- **Smoothing:** rm noise.
- **Aggregation:** summarise, build data cube.
- **Generalisation:** climb concept hierarchy (street → city → country).
- **Normalisation:** scale into a small range.
- **Attribute construction:** build new attrs from given ones.

Why normalise

- Stops large-range attrs dominating small-range ones.
- Required by distance-based methods: KNN, K-means, ANN.

The 3 normalisation formulas (memorise exactly)

1. Min-max → maps to $[new_min, new_max]$:

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

2. Z-score (zero-mean) → use when min/max unknown or outliers present:

$$v' = \frac{v - \bar{A}}{\sigma_A} \quad \bar{A} = \text{mean}, \sigma_A = \text{std deviation}$$

3. Decimal scaling:

$$v' = \frac{v}{10^j} \quad j = \text{smallest int s.t. } \max(|v'|) < 1$$

Variant of 2: z-score w/ mean absolute deviation s_A , more robust to outliers than σ_A (deviations are not squared):

$$s_A = \frac{1}{n} \sum_i |v_i - \bar{A}| \quad v' = \frac{v - \bar{A}}{s_A}$$

Error in your class notes verified/added : z-score is printed there as $v' = (v - min)/\sigma$. That is **wrong**. It is $(v - \text{mean})/\sigma$. Using min makes every $v' \geq 0$, which defeats the whole point of a zero-mean score.

(d) Data reduction 5

- Smaller volume, **same (or nearly same) analytical result.**
- **Strategies:**
 - **Sampling** (below)
 - **Dimensionality reduction:** PCA, SVD, wavelet
 - **Feature subset selection**
 - **Aggregation** / data cube
 - **Discretisation** + concept hierarchy
 - **Compression:** lossy / lossless
- **Han & Kamber's 3-way split** NEW (**safer skeleton for a 5-mark answer**):
 - **Dimensionality reduction:** wavelet transform, PCA, attribute subset selection. *Fewer attributes.*
 - **Numerosity reduction:** replace the data by a smaller representation. *Fewer tuples.*
 - *Parametric:* fit a model, store only its parameters → regression, log-linear models.
 - *Non-parametric:* store a reduced form directly → histograms, clustering, sampling, data cube aggregation.
 - **Data compression:** lossless (exact rebuild) vs lossy (approximation). The other two are themselves forms of compression.

Sampling types

- **Simple random:** equal prob for each item.
- **W/o replacement:** item removed once picked.
- **W/ replacement:** not removed, same item can repeat.
- **Stratified:** split into partitions, sample each. Best for skewed data.

Feature subset selection

- Rm **redundant** features (purchase price + sales tax paid).
- Rm **irrelevant** features (student ID for predicting GPA).
- **Techniques**:
 - *Brute force*: try all subsets.
 - *Embedded*: selection happens inside the algorithm.
 - *Filter*: select before the algorithm runs.
 - *Wrapper*: use algorithm as a black box to score subsets.

Discretisation (feeds Ch 3 decision trees)

- Continuous $\rightarrow$ small no. of interval labels. Many algorithms need discrete input.
- **Unsupervised** (no class labels):
 - Equal-width, equal-frequency, **k-means** (cluster values, 1 bin per cluster).
- **Supervised** (uses class labels): **entropy-based**.
 - $H(p_1, \dots, p_s) = \sum_i p_i \log_2(1/p_i) = -\sum_i p_i \log_2(p_i)$
same quantity as *Info(D)* in Ch 3, §3.2 – written both ways in the literature
 - Split S at boundary T :
 $E(S, T) = \frac{|S_1|}{|S|} Ent(S_1) + \frac{|S_2|}{|S|} Ent(S_2)$
 - Pick T **minimising** E ; recurse until
 $Ent(S) - E(T, S) \leq \delta$.
- Also: **mapping to a new space** (Fourier, wavelet transform).

(e) PCA for dimensionality reduction 10 4

71 Shr 10, 76 Ch 4 numerical Numerical worked in the companion.

Key idea

- Find a **projection capturing the largest variance** in the data.
- New axes = eigenvectors of the **covariance matrix**, ordered by eigenvalue.
- Eigenvalue λ_i = variance carried by principal component i .
- Keep the top r components $\rightarrow$ drop the rest w/ little info loss.
- Components are **orthogonal** (uncorrelated).

Proportion of variance explained

$$\frac{\sum_{i=1}^r \lambda_i}{\sum_{i=1}^m \lambda_i} = \frac{\lambda_1 + \dots + \lambda_r}{\lambda_1 + \dots + \lambda_m}$$

- Denominator = trace of covariance matrix = total variance.

Curse of dimensionality 70 Ch 5

- As dims $\uparrow$, data becomes increasingly **sparse** in the space it occupies.
- Density + distance lose meaning $\rightarrow$ breaks clustering + outlier detection.
- All points start looking equidistant.
- **Avoid by**: dimensionality reduction (PCA/SVD), feature subset selection, feature construction, domain knowledge.

PCA process: 5 steps (DBK's sequence)

1. **Subtract the mean** from each dimension $\rightarrow$ zero-mean data.
2. **Compute the covariance matrix**.
3. **Compute eigenvalues + eigenvectors** of that matrix.
4. **Choose components, form feature vector**: sort λ descending, keep top r , put their eigenvectors in columns.
5. **Derive new data**: $\text{FinalData} = \text{RowFeatureVector} \times \text{RowZeroMeanData}$.

Notes

- Eigenvectors exist only for **square** matrices.
- An $m \times m$ matrix gives m of them.
- Eigenvectors of a **symmetric** matrix are mutually perpendicular.
- Normalise eigenvectors to unit length.
- Highly correlated dims $\rightarrow$ few large $\lambda \rightarrow$ PCA helps a lot.
 Uncorrelated dims $\rightarrow r \approx m \rightarrow$ **PCA does not help**.

Error in your class notes verified/added : the PCA section there lists "construct a neighbourhood graph" and "compute shortest-path / geodesic distances". Those steps belong to **Isomap**, a non-linear method, **not** PCA. PCA is purely linear: covariance $\rightarrow$ eigenvectors.

2.3 OLAP & Multidimensional Data Analysis

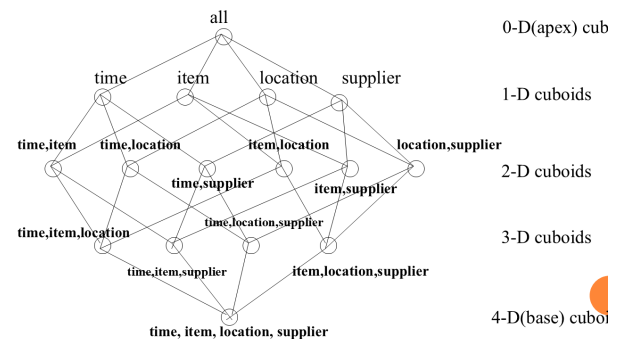
Explain typical OLAP operations over a multidim DW. Differentiate OLAP vs OLTP. 6+4 10 5+5 2+5+3

TOP 13/23 81 Bh, 79 Bh, 75 Ash, 74 Ch, 73 Ch, 72 Ch, 70 Asa

Multidimensional model

- **Dimension:** perspective the org keeps records by. Eg time, item, location, branch.
- **Measure:** the numeric quantity analysed. Eg units_sold, dollars_sold.
- **Fact table:** stores measures + FKs to dimension tables.
- **Data cube:** lets you view measures against a set of dims.
- **Cuboid:** the cube at one level of summarisation.
 - **Base cuboid** = lowest level, all n dims.
 - **Apex cuboid** = 0-D, highest summary, denoted all.
 - Lattice of cuboids = the full cube. n dims $\rightarrow 2^n$ cuboids.
- Schemas: **star**, **snowflake**, **fact constellation** $\rightarrow$ defined in Ch 1.
- **Concept hierarchy:** the levels of abstraction inside 1 dim, eg street < city < province < country.
 - This is **what roll-up and drill-down actually climb**.
 - Supplied manually by users/experts, or generated automatically from the data distribution.

Cube: A Lattice of Cuboids



3 kinds of measure, by aggregate function

- **Distributive:** partition the data, aggregate each part, aggregate the results $\rightarrow$ same answer. `sum()`, `count()`, `min()`, `max()`.
- **Algebraic:** computable from a bounded no. of distributive functions. `avg()` = `sum()/count()`, `min_N()`, `standard_deviation()`.
- **Holistic:** no constant-size sub-aggregate suffices. `median()`, `mode()`, `rank()`. **Expensive** $\rightarrow$ usually estimated.

The 5 OLAP operations (the core answer)

- **Roll-up (drill-up):** summarise by climbing a hierarchy or reducing a dim. Eg city $\rightarrow$ country; or drop *time* entirely.
- **Drill-down (roll-down):** reverse of roll-up. Summary $\rightarrow$ detail, or introduce a new dim. Eg quarter $\rightarrow$ month.
- **Slice:** select on **one** dim $\rightarrow$ sub-cube. Eg *time* = Q1.
- **Dice:** select on **two or more** dims. Eg *location* in {Kathmandu, Pokhara} and *time* in {Q1,Q2}.
- **Pivot (rotate):** rotate axes to give an alternative presentation. 3-D $\rightarrow$ series of 2-D planes.
- **Others:** **drill-across** (across ≥ 2 fact tables), **drill-through** (down to the back-end relational tables).

OLAP vs OLTP (asked 4 times)

- **Users:** OLTP clerks / IT · OLAP knowledge workers.
- **Function:** OLTP day-to-day ops · OLAP decision support.
- **DB design:** OLTP application-oriented, normalised (ER) · OLAP subject-oriented, de-normalised (star/snowflake).
- **Data:** OLTP current, detailed, isolated · OLAP historical, summarised, multidim, consolidated.
- **Usage:** OLTP repetitive · OLAP ad-hoc.
- **Access:** OLTP read/write, index on primary key · OLAP lots of scans.
- **Unit of work:** OLTP short simple txn · OLAP complex query.
- **Records accessed:** tens · millions.
- **Size:** 100 MB-GB · 100 GB-TB.
- **Metric:** txn throughput · query throughput / response.

OLTP vs. OLAP

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date, detailed, relational, isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

OLAP server types (detail in Ch 1, DW architecture)

- **ROLAP**: relational back end. Scales best.
- **MOLAP**: multidim array store. Fastest, wastes space on sparse data.
- **HOLAP**: hybrid. Detail relational + aggregates multidim.

Base & apex cuboid, slice & dice, roll-up & drill-down w/ example 2+3+3+3

76 Ash Answer = the definitions above + 1 worked example each. Cuboid-count and schema-drawing questions are worked in the Numerical companion
78 Bh, 76 Ch.

2.4 Various Similarity Measures**How is similarity/dissimilarity calculated? Properties of a distance metric.** 3 10 4

HOT 7/23 80 Bh, 80 Ba, 78 Bh, 76 Ch, 76 Ash, 71 Ch, 70 Asa **Almost always numerical** → companion.

Similarity vs dissimilarity

- **Similarity**: how alike 2 objects are. *Higher* = more alike. Usually $[0, 1]$.
- **Dissimilarity / distance**: how different. *Lower* = more alike. Min = 0, upper limit varies, often $[0, \infty]$.
- Used for: clustering, KNN, outlier detection, search, recommendation, near-duplicate detection.

Distance metric: 4 properties (71 Ch asks exactly this, 10)

- 1. Non-negativity**: $d(x, y) \geq 0$.
- 2. Identity**: $d(x, y) = 0$ iff $x = y$.
- 3. Symmetry**: $d(x, y) = d(y, x)$.
- 4. Triangle inequality**: $d(x, y) \leq d(x, z) + d(z, y)$.
 - Triangle inequality → guarantees the function is *well behaved*: the direct connection is the shortest.
 - A measure satisfying all 4 = a **metric**. Minkowski distances are metrics for $r \geq 1$.
 - **Subtle point worth 1 mark**: non-negativity is *implied* by the other 3, so strictly only 3 are independent.
 - **Weighted Euclidean** (attrs of unequal importance):

$$d(i, j) = \sqrt{\sum_k w_k (x_{ik} - x_{jk})^2}.$$

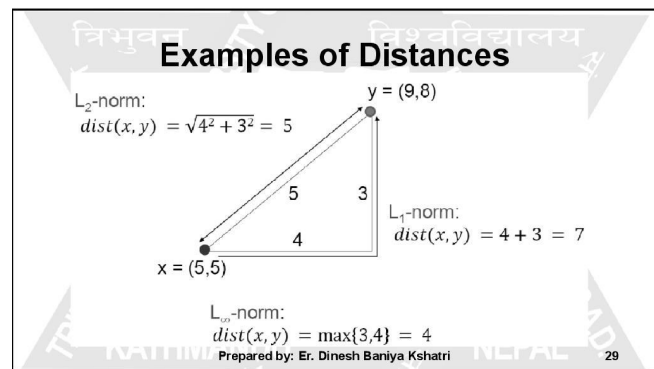
Minkowski family: the one formula that generates the rest

$$\text{dist}(p, q) = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{1/r}$$

- n = no. of dims, r = parameter, $r \geq 1$. **Do not confuse r with n .**
- **Notation clash:** H&K call the parameter h and reserve p for the no. of attrs; other books write L_p where their p is this r . **Define your symbols.**
- $r = 1 \rightarrow$ **Manhattan** (L_1 , city block, taxicab). Binary case = **Hamming distance**.
- $r = 2 \rightarrow$ **Euclidean** (L_2).
- $r \rightarrow \infty \rightarrow$ **Supremum** ($L_{\max}$, L_∞ , Chebyshev, chessboard) = max difference in any single coordinate.

Written out

- Euclidean: $\sqrt{\sum_k (p_k - q_k)^2}$
- Manhattan: $\sum_k |p_k - q_k|$
- Supremum: $\max_k |p_k - q_k|$



Minkowski Distance – Example

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

I.1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

I.2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

I.3	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Prepared by: Er. Dinesh Baniya Kshatri

Binary attributes: SMC vs Jaccard 2+2+2

Build a 2×2 contingency count for objects i, j . **Two notations exist, know both:**

- $q = M_{11}$ = attrs where both = 1 $t = M_{00}$ = attrs where both = 0
- $r = M_{10} = i=1, j=0$ $s = M_{01} = i=0, j=1$
- Total $p = q + r + s + t$. **H&K use q, r, s, t ; Tan and DBK use M_{11} etc.**

$$d = \frac{r + s}{q + r + s + t} \quad d = \frac{r + s}{q + r + s} \quad SMC = \frac{q + t}{q + r + s + t} = 1 - d_{sym}$$

symmetric asymmetric

- **Only difference: Jaccard/asymmetric drop t (the 0-0 matches).**
- **SMC** $\rightarrow$ **symmetric** binary (a 0-0 match is informative).
- **Jaccard** $\rightarrow$ **asymmetric** binary (0-0 meaningless: 2 people both lacking a rare disease says nothing).
- Sets form: $J(A, B) = |A \cap B| / |A \cup B|$.
- **Read the verb.** "Find the Jaccard coefficient" wants J ; "find the dissimilarity" wants $d = 1 - J$.

SMC vs. Jaccard Example

$p = 1000000000$
 $q = 0000001001$

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)
 $M_{10} = 1$ (the number of attributes where p was 1 and q was 0)
 $M_{00} = 7$ (the number of attributes where p was 0 and q was 0)
 $M_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = (0 + 7) / (2 + 1 + 0 + 7) = 0.7$

$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0 / (2 + 1 + 0) = 0$

Prepared by: Er. Dinesh Baniya Kshatri

Nominal attributes

$$d(i, j) = \frac{p - m}{p} \quad \text{sim}(i, j) = 1 - d(i, j) = \frac{m}{p}$$

- m = no. of attrs where i and j are in the **same state**, p = total attrs.
- Alternative: encode each of the M states as an **asymmetric binary attr**, then use Jaccard.

Ordinal attributes

1. Replace value x_{if} by its **rank** $r_{if} \in \{1, \dots, M_f\}$.
2. Normalise rank to $[0, 1]$ so every attr has equal weight:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

3. Treat z_{if} as **numeric** → use Euclidean / Manhattan.
- Eg {fair, good, excellent}, $M_f = 3 \rightarrow$ ranks 1,2,3 → $z = 0, 0.5, 1$.

Cosine similarity 3

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}$$

- $d_1 \cdot d_2$ = dot product, $\|d\| = \sqrt{\sum d_k^2}$ = vector length.
- Range $[0, 1]$ for non-negative vectors.
- Aligned → angle $0^\circ \rightarrow \cos = 1$. Orthogonal (no common terms) → $90^\circ \rightarrow \cos = 0$.
- **Ignores magnitude, measures direction only** → ideal for documents, where doc length shouldn't matter.

Which measure for which data (picking wrong = losing the marks)

Data	Use
Numeric, scale matters	Euclidean / Manhattan / Minkowski
Numeric, only direction matters (documents)	Cosine
Symmetric binary	SMC
Asymmetric binary, sets	Jaccard
Nominal / categorical	Simple mismatch $d = (p - m)/p$, m = no. of matches, p = total attrs
Strings	Hamming (equal length) / Levenshtein edit distance
Testing linear relationship	Correlation coefficient

Data matrix vs dissimilarity matrix

- **Data matrix** (n objects $\times p$ attrs): stores the objects themselves. **Two-mode** (rows = objects, cols = attrs).
- **Dissimilarity matrix** ($n \times n$): stores $d(i, j)$ for every pair. **One-mode** (both axes are objects).
- Properties: $d(i, i) = 0$, $d(i, j) = d(j, i) \rightarrow$ only the **lower triangle** is written.
- Most clustering + KNN algorithms run on the dissimilarity matrix.

Mixed attribute types

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

- $\delta_{ij}^{(f)} = 0$ if the attr is missing for either object, or it is asymmetric binary and both are 0. Else $\delta = 1$.
- $d_{ij}^{(f)}$ computed per attr type, each scaled to $[0, 1]$ (numeric: divide by the range).

Correlation coefficient

$$\text{corr}(x, y) = \frac{\sum_i (x_i - \mu_x)(y_i - \mu_y)}{\sqrt{\sum_i (x_i - \mu_x)^2} \sqrt{\sum_i (y_i - \mu_y)^2}} = \frac{s_{xy}}{s_x s_y}$$

- = cosine similarity of the **mean-centred** vectors.
- Range $[-1, 1]$: +1 positive, -1 negative, 0 no correlation.
- **Covariance** $s_{xy} = \frac{1}{n-1} \sum_k (x_k - \bar{x})(y_k - \bar{y})$.
- Correlation = scaled (unit-free) version of covariance.

2.5 Basic Statistical Descriptions of Data

NEW Han & Kamber 3rd ed §2.2. **Never asked in the 23 papers**, so no tier chip. Added because PC's 2025–26 Unit 2 handout and its numerical companion are built entirely on it, and because the IQR outlier rule below is the statistical basis of the boxplot test used in Ch 6. **Learn the two boxed formulas; skim the rest.**

Measures of central tendency

- **Mean** $\bar{x} = \frac{1}{n} \sum_i x_i$ (sample), $\mu = \frac{1}{N} \sum x$ (population).
- **Weighted mean** $\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$.
- **Trimmed mean**: chop the extreme values first → kills outlier sensitivity. Chop at most 2% per end.
- **Median**: middle value if n odd, mean of the middle two if n even.
- **Median is holistic** → needs a full sort → expensive on a large table. (Same holistic/distributive/algebraic split as the cube measures, §2.3.)
- **Mode**: most frequent value. Unimodal / bimodal / trimodal / multimodal. No mode if every value occurs once.
- **Midrange** = $\frac{\max + \min}{2}$. Cheap: 1 pass, $\max()$ and $\min()$.

Median of grouped data (intervals, not raw values):

$$\text{median} = L_1 + \left(\frac{N/2 - (\sum \text{freq})_l}{\text{freq}_{\text{median}}} \right) \times \text{width}$$

L_1 = lower boundary of the median interval, N = total count, $(\sum \text{freq})_l$ = cumulative frequency of all intervals *below* it, $\text{freq}_{\text{median}}$ = its own frequency, width = its width.

Locate the interval first: the one where cumulative frequency first reaches $N/2$.

Empirical relation, unimodal and moderately skewed:

$$\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$$

$$\Rightarrow \text{mode} \approx 3\text{median} - 2\text{mean}$$

Skew, read off mean vs median vs mode

- **Symmetric**: $\text{mean} = \text{median} = \text{mode}$.
- **Positively skewed** (right tail): $\text{mode} < \text{median} < \text{mean}$.
- **Negatively skewed** (left tail): $\text{mean} < \text{median} < \text{mode}$.
- **Mean is dragged toward the tail**, median is not. That is why median is the robust one.

Measures of dispersion

- **Range** = $\max - \min$.
- **Quartiles**: Q_1 = 25th percentile, Q_2 = median, Q_3 = 75th percentile.
- **IQR** = $Q_3 - Q_1$ = spread of the middle half.
- **Variance** $s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$, **std deviation** $s = \sqrt{s^2}$.
- Population forms use $\frac{1}{N}$ and σ .

Two quartile conventions (they disagree, say which you used)

- **Positional (Han & Kamber, the syllabus book):** for $n = 12$ take the 3rd, 6th and 9th sorted values directly.
- **Interpolated (PC's class notes):** position of $Q_k = \frac{k(n+1)}{4}$, then linear interpolation between the two neighbouring values.
- Both are marked correct. **The outlier verdict comes out the same either way** → see Ch 2 Numerical, M7.

Five-number summary and boxplot

- **Five-number summary**, in this order:
Minimum, Q_1 , Median, Q_3 , Maximum.
- **Boxplot:** box spans Q_1 to Q_3 (box length = IQR), line inside marks the median, two **whiskers** extend to min and max, outliers plotted as individual points.

IQR outlier rule (the one to memorise): a value is a suspected outlier if

$$x < Q_1 - 1.5 IQR \quad \text{or} \quad x > Q_3 + 1.5 IQR$$

Beyond $3 IQR = \text{extreme outlier}$. **Used again in Ch 6** as the boxplot test.

Graphic displays

- **Histogram:** $x = \text{value / bin}$, $y = \text{frequency}$.
- **Quantile plot:** each x_i against f_i , the fraction of data at or below it.
- **Quantile-quantile (q-q) plot:** quantiles of one distribution against the corresponding quantiles of another → compares 2 datasets.
- **Scatter plot:** each pair as a point → shows correlation and clusters.
- **Histogram often tells more than a boxplot:** two datasets can share a boxplot and have completely different histograms.

2.6 PYQ Mapping

Tiers: **TOP /23** = 9+/23 **HOT /23** = 5-8 **PIN /23** = 1-4 but ≥ 8 marks. **N** marks a question fully solved in the Ch 2 Numerical companion.

2.1 Data Types and Attributes

1. **PIN 1/23** "What is data mining? Explain different data types of attributes in a dataset." 10 71 Shr

2.2 Data Pre-processing

2. **TOP 12/23** **Why preprocess + major tasks / methods.**
 - "What is Data Pre-Processing? Briefly explain the major tasks performed." 2+7 81 Ba
 - "Why data preprocessing is necessary? Explain methods to maintain data quality." 4+4 75 Ch
 - "Why is it required? Explain approaches of data cleaning." 5+5 72 Ka
3. **PIN 1/23** "Measuring elements of data Quality? Data transformation by normalization w/ example." 2+6 73 Shr
4. **N** "Use the following methods to normalize 200, 300, 400, 600, 1000" (min-max, z-score, decimal scaling). 2+2+2 78 Bh
5. **PIN 2/23** Handling missing values. 5 74 Ash 2 74 Ch
6. **PIN 1/23** Issues in Data Integration. 5 75 Ash
7. **PIN 1/23** "What is 'curse of Dimensionality'? How can it be avoided?" 5 70 Ch **paper typo: "curse"**
8. **PIN 1/23** "Discuss the impact of noisy data in data mining." 5 70 Ch
9. **PIN 1/23** "How can PCA be used for dimensionality reduction?" 10 71 Shr
10. **N** "Find the principal components + proportion of total variance" (3×3 covariance matrix). 4 76 Ch
11. **PIN 3/23** Short notes: data smoothing 76 Ash · data normalization 75 Ch · data transformation 73 Shr, 72 Ka. 3 4

2.3 OLAP & Multidimensional Data Analysis

12. **TOP 10/23** **OLAP operations + OLAP vs OLTP.**
 - "Explain typical OLAP operations...Differentiate OLAP and OLTP tools." 6+4 79 Bh
 - "How do you perform analysis of multidimensional data? Explain w/ OLAP." 10 72 Ch
 - "Discuss Data Integration issues. Describe OLAP + operations w/ example." 5+5 75 Ash
 - "Approaches to handle missing data? OLAP + operations? OLAP vs OLTP." 2+5+3 74 Ch
13. **PIN 1/23** "Dimensional data? Base & apex cuboid? Slicing & Dicing? Roll Down/Up? Give example." 2+3+3+3 76 Ash

14. **N** Star schema for a 4-dim DW + which OLAP ops list total charge. 3+3 78 Bh
15. **N** Snowflake schema for a 5-dim DW + OLAP ops for count per state per year. 3+3 76 Ch
16. **PIN 3/23** Short notes: OLAP Operations 74 Ash, 73 Shr · OLAP and OLTP 75 Ch. 4

2.4 Various Similarity Measures

17. **N** Cosine sim (Obj 2,4) + Euclidean (1,3) and (1,4), 4-attribute object table. 3+3+3 80 Bh
18. **N** Jaccard (asymmetric) + symmetric-binary dissimilarity + SMC + cosine of 2 doc vectors. 2+2+2+1 80 Ba
19. **N** Manhattan + Supremum distance matrix · cosine of 2 vectors · Jaccard + SMC of 2 binary vectors. 2+1+2 76 Ch
20. **PIN 1/23** "Properties a Distance Metric needs to support w.r.t. Minkowski's distance." 10 71 Ch
21. **PIN 1/23** "List types of similarity measures + their application areas." 3 80 Ba
22. **PIN 2/23** Short notes: Minkowski Distance 78 Bh · similarity measures between tuples 76 Ash. 4 3

Papers added by the 2026-09 merge

The six papers of `DM4.1BCT.pdf` (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. verified/added

- What is performed during preprocessing? + techniques to handle missing values. 4+4 82 Bh
- **Curse of dimensionality** in data mining. 6 81 Bh
- Why are similarity measures important? + the different measures w/ examples. 2+6 81 Bh
- **N** Bin means at depth 3 + min-max + z-score + decimal scaling of 35, on 27 values. 6 79 Ba
- Nominal and ordinal attributes + handling missing and noisy data. 6 79 Ba
- Preprocessing + **data sampling** + dimensionality reduction w/ example. 2+4+4 73 Ch
- How does real data become **asymmetric**? Symmetric vs asymmetric data. 5 73 Ch
- What is dimensionality reduction and why does it matter? 5 70 Asa
- Properties of a distance metric (its second half is §3.4). 10 70 Asa
- Different data types w/ examples 69 Ch · short note **OLAP cubes** 70 Asa. 5 5

Coverage check

- All Ch 2 PYQs covered. 8 are numerical → fully solved in the companion.
- **3 errors in your source notes corrected here** verified/added :
 - Z-score written as $(v - \min)/\sigma \rightarrow$ must be $(v - \text{mean})/\sigma$.
 - Equal-width bin width $W = (A - B)/N \rightarrow$ must be $(B - A)/N$.
 - PCA section contaminated w/ Isomap steps (neighbourhood graph, geodesic distances).
- **Gap filled by research** verified/added : similarity-measure formulas were absent from the typed notes (they say only "Refer Class Note"), so they were taken from DBK Lecture-2 and then **verified against Han & Kamber 3rd ed** (the syllabus reference book).

Sources used for this chapter

- **Han & Kamber, 3rd ed** (full textbook PDF): ch 2 attribute types + all proximity measures, ch 3 preprocessing + normalisation, ch 4 data cube + OLAP. **Primary authority.**
- **DBK Lectures 1–4** (Thapathali class slides): attribute types, binning example, discretisation, distance/similarity, PCA process, data cube + concept hierarchies.
- `data mining_/Chapter 1_2.pdf` (Bhupesh K. Mishra), `Chapter3_0.ppt`, `Chapter 2_Data Warehouse.ppt`.
- **Notes by PC** (Er. Prakash Chandra Prasad, Asst. Prof., IOE Pulchowk), dated 2025–26: `Unit 2.docx`, `unit 2 Numerical.docx`, `Redundancy and Correlation Analysis.docx`. Source of the new §2.5 and of the χ^2 procedure in §2.2(b). **Newest teaching material available**, so treat §2.5 as insurance even though no past paper has asked it.
- Where sources disagree, **Han & Kamber 3rd ed wins** and the clash is flagged inline.

3. Classification

12 hrs · in 23/23 papers · 7 topics, 3 TOP tier · largest chapter in the syllabus · numericals → Ch 3 companion.

Weight of the chapter:

- Decision Tree Classifier **TOP 9/23** → nearly always a full numerical.
- Bayesian Classifier **TOP 8/23**
- Issues: Overfitting, Validation, Model Comparison **TOP 7/23** → **confusion-matrix numerical appears in 8 papers.**
- ANN Classifier **HOT 5/23** · Rule Based **HOT 4/23**
- Basics **PIN 3/23** · Nearest Neighbor **PIN 2/23**

3.1 Basics and Algorithms

Why is classification a supervised learning method? **2** **10**

PIN 4/23 82 Bh, 80 Bh, 74 Ch, 71 Shr

Classification

- Predicts **categorical** class labels (discrete, nominal).
- **Supervised**: the training tuples come w/ a known **class label**. The model learns from labelled answers.
- vs **unsupervised** (clustering): no labels given, groups discovered. → Ch 5
- vs **prediction / regression**: predicts a *continuous* value, not a label.

2-step process (**this is the block diagram Q**)

1. **Learning / model construction** (training)
 - Input: **training set**, each tuple w/ its class label.
 - Output: the model, as rules / tree / formula.
2. **Classification / model usage** (testing then deployment)
 - Estimate accuracy on an **independent test set**.
 - **Test set must be separate from the training set**, else the accuracy is optimistic (the model has memorised).
 - If accuracy acceptable → classify new, unlabelled data.

Where each classifier fits

- **Eager learners** build the model *before* seeing a query: decision tree, Bayes, ANN, rule-based, SVM.
- **Lazy learners** store the data and act only at query time: KNN, case-based.

Applications

- Credit / loan approval, fraud detection, medical diagnosis, target marketing, churn prediction, spam filtering.

Answer skeleton **10**

1. Definition + “supervised because labels are given”.
2. **Draw the 2-step block diagram** (training → model → testing → deployment).
3. Name the classifiers, 1 line each.
4. Close w/ the 2–3 metrics used to judge one.

3.2 Decision Tree Classifier

What is a decision tree? How is information gain used for attribute selection? 8 10 4+6

TOP 13/23 82 Bh, 81 Bh, 79 Bh, 78 Bh, 76 Ch, 76 Ash, 75 Ch, 74 Ch, 73 Ch, 73 Shr, 72 Ka, 71 Ch, 70 Asa **Almost always a full tree-building numerical**
→ companion.

Structure

- Flow-chart-like **inverted tree**.
- **Root node**: topmost, the first test.
- **Internal node**: a test on one attribute.
- **Branch**: an outcome of that test.
- **Leaf node**: holds a **class label**.
- Path root → leaf = one classification rule.

8.2 Decision Tree Induction 331

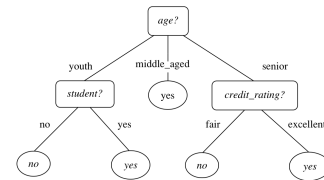


Figure 8.2 A decision tree for the concept *buys.computer*, indicating whether an *AllElectronics* customer is likely to purchase a computer. Each internal (nonleaf) node represents a test on

Why popular

- No domain knowledge or parameter setting needed.
- Handles high-dimensional data.
- **Human-readable** → easy to convert to IF-THEN rules.
- Fast; good accuracy in most cases.

Hunt's algorithm 72 Ka 10 → the ancestor of all of them

Let D_t = training records reaching node t , $y = \{y_1 \dots y_c\}$ the class labels.

1. If all records in D_t belong to the same class $y_t \rightarrow t$ is a **leaf** labelled y_t .
 2. If D_t is empty → leaf labelled w/ the **majority class of the parent**.
 3. If D_t has records of more than 1 class → pick an **attribute test**, split D_t into subsets, **recurse** on each subset.
- Descendants: **ID3** → **C4.5** → C5.0 (Quinlan), **CART**, SLIQ, SPRINT.

Generic induction algorithm

- **Greedy, top-down, recursive, divide-and-conquer**.
- Greedy = picks the locally best attribute at each node, never backtracks.
- **Stopping conditions**:
 - all tuples at the node share 1 class, or
 - no attributes left → leaf = majority class, or
 - no tuples left → leaf = majority class of parent.

Test conditions per attribute type 72 Ka asks this

- **Binary**: 2 outcomes, forced.
- **Nominal**
 - **Multi-way**: 1 branch per value.
 - **Binary**: split values into 2 subsets. $2^{v-1} - 1$ ways.
- **Ordinal**: same as nominal, but a grouping must **not break the order**.
- **Continuous**
 - **Binary**: test $A \leq t$ vs $A > t$.
 - **Multi-way**: discretise into ranges first.
 - **Split point**: sort values, try the **midpoint** of each adjacent pair, $\frac{a_i + a_{i+1}}{2}$; v values → $v - 1$ candidate splits. Pick the one w/ minimum $Info_A(D)$.

The 3 attribute selection measures (memorise all three)

1. Information gain ID3 maximise

$$Info(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times Info(D_j)$$

$$Gain(A) = Info(D) - Info_A(D)$$

- $p_i = |C_{i,D}|/|D|$ = fraction of class i . $Info(D)$ = **entropy**, in bits.
- Pick the attribute w/ the **highest** $Gain$.
- **Bias**: prefers attrs w/ **many distinct values** (a unique ID gives $Info_A = 0$, so maximal gain, yet is useless).

2. Gain ratio C4.5 maximise

$$SplitInfo_A(D) = - \sum_{j=1}^v \frac{|D_j|}{|D|} \log_2\left(\frac{|D_j|}{|D|}\right)$$

$$GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)}$$

- Normalises the gain → **fixes the many-values bias**.
- Unstable as $SplitInfo \rightarrow 0$; constrain $Gain$ to be at least the average.

Overfitting + pruning 75 Ch 4

- Tree grows branches that fit **noise and outliers** → poor accuracy on new data.
- **Pre-pruning** (early stopping)
 - Halt splitting at a node; make it a leaf w/ the majority class.
 - Stop if: gain below a threshold, too few tuples, purity high enough.
 - **Hard to pick the threshold** → too high underfits, too low overfits.
- **Post-pruning** (grow full, then cut) → **preferred**
 - Remove subtrees from a fully grown tree, replace w/ a leaf.
 - CART uses *cost complexity* pruning w/ a separate prune set.
 - C4.5 uses *pessimistic* pruning, no extra data needed.
- Other fixes: more training data, cross-validation, ensembles.

3. Gini index CART minimise

$$Gini(D) = 1 - \sum_{i=1}^m p_i^2$$

$$Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2)$$

$$\Delta Gini(A) = Gini(D) - Gini_A(D)$$

- **Gini considers only BINARY splits** → for v values, try all $2^{v-1} - 1$ subset splits and keep the best.
- Pick the attribute w/ the **largest** $\Delta Gini$ (equivalently smallest $Gini_A$).
- **Bias**: toward attrs w/ many values, and toward equal-sized balanced partitions.

Reading the measures

- All 3 measure **impurity**. Pure node → $Info = 0$, $Gini = 0$.
- Max impurity: 2 classes 50/50 → $Info = 1$, $Gini = 0.5$.

Advantages

- Simple, fast to build and to apply.
- Self-explanatory, converts directly to rules.
- No normalisation or scaling needed.
- Handles both nominal and numeric attrs, and irrelevant attrs.

Disadvantages

- **Greedy** → locally optimal, not guaranteed globally optimal.
- Prone to **overfitting** without pruning.
- Unstable: a small data change can restructure the whole tree.
- Axis-parallel splits only → struggles w/ diagonal boundaries.
- Biased toward attrs w/ many values (unless gain ratio used).

3.3 Rule Based Classifier

How does a Rule Based Classifier work? Explain with example. 7 2+8 9

HOT 5/23 80 Bh, 76 Ch, 74 Ash, 70 Ch, 69 Ch

Form

- Model = a set of **IF-THEN** rules.

-

IF (age = youth) AND (student = yes) THEN buys_computer =

- **Antecedent** (LHS, the condition) → **Consequent** (RHS, the class).
- A rule **covers** a tuple if the antecedent holds; the rule is then said to be *triggered*.

Rule quality measures 70 Ch asks Accuracy + Laplace

verified/added

- n_{covers} = tuples covered by R , $n_{correct}$ = those it labels correctly, $|D|$ = size of the data set, k = no. of classes.

$$Coverage(R) = \frac{n_{covers}}{|D|} \quad Accuracy(R) = \frac{n_{correct}}{n_{covers}}$$

$$Laplace(R) = \frac{n_{correct} + 1}{n_{covers} + k} \quad m\text{-estimate}(R) = \frac{n_{correct} + k p}{n_{covers} + k}$$

- **Why Laplace**: plain accuracy rates a rule covering 1 tuple correctly as 1.0, which is meaningless. Laplace pulls small-coverage rules toward the prior, so a rule must cover *enough* tuples to score well.
- Eg 1 tuple, 1 correct, 2 classes: $Accuracy = 1/1 = 1.0$ but $Laplace = 2/3 = 0.67$.

Rule extraction from a decision tree

- 1 rule per **root-to-leaf path**. Each split condition ANDed; leaf = consequent.
- Mutually exclusive + exhaustive by construction.
- Rules can then be **pruned** individually, which may break exclusivity → then class-based ordering is applied.

Conflict resolution 76 Ch 2+4

- **Why needed**: a tuple may trigger **several rules predicting different classes**, or **no rule at all**.
- **Size ordering** (rule antecedent size): the *most specific* rule wins, ie the one w/ the most attribute tests. Intuition: more conditions = more evidence.
- **Class-based ordering**: classes sorted by decreasing prevalence (or by misclassification cost). All rules of one class come as a block.
- **Rule-based ordering (decision list)**: rules ranked by quality (accuracy, coverage, size) by the expert or the algorithm. **First rule that fires wins**.
- **No rule fires** → use a **default rule**, predicting the majority class of the tuples not covered by any rule. Always placed last.

Two useful properties

- **Mutually exclusive**: no 2 rules fire on the same tuple → no conflict possible.
- **Exhaustive**: every possible tuple is covered → no default rule needed.
- Rules extracted from a decision tree are **both** by construction.

Sequential covering (direct rule induction) 70 Ch asks CN2

verified/added

- Learns rules **directly from the data**, one class at a time.
- Algorithms: **CN2**, **RIPPER**, **AQ**, **FOIL**.
- Steps:
 1. Pick a class. Learn 1 rule that covers many of its tuples and few others.
 2. **Remove** the tuples that rule covers.
 3. Repeat on what remains until a stopping criterion is met.
- **CN2** specifically: general-to-specific beam search; starts from the empty antecedent and repeatedly appends the test that most improves the rule, scored by **entropy** and checked by a **likelihood-ratio significance test**. Produces an ordered rule list (decision list) + a default rule.

Decision tree classifier vs rule-based classifier 8

69 Ch “How is decision tree classifier different than rule based classifier?” **Note they are relatives, not rivals:** every tree converts to a rule set, so say that first and then give the differences. verified/added

	Decision tree	Rule-based
Model form	One connected tree ; internal node = attribute test, leaf = class.	An unordered set or an ordered list of IF-THEN rules.
How it is built	Divide and conquer : recursively split the whole data set on the best attribute (Hunt’s algorithm, ID3, C4.5, CART).	Separate and conquer (sequential covering): learn one rule, remove the tuples it covers, repeat on the remainder (CN2, RIPPER).
Coverage	Partitions the space exactly once : rules are automatically mutually exclusive and exhaustive , so every tuple lands in exactly one leaf.	Rules may overlap or leave gaps → needs a conflict-resolution strategy (size / class / rule ordering) and a default rule .
Attribute reuse	A test near the root constrains every rule below it, so an attribute chosen early is forced on all branches.	Each rule is grown independently, so rules need share no condition at all .
Pruning	Prunes <i>subtrees</i> ; removing a node changes every path through it.	Prunes <i>individual rules</i> , and may drop conditions from one rule without touching the rest. More flexible, but can break mutual exclusivity .
Replicated subtree problem	Suffers from it: the same subtree is duplicated under several branches when the concept is disjunctive.	Immune : one rule expresses the disjunct once.
Readability	Whole model visible in one diagram; easy to explain end to end, but a deep tree becomes unreadable.	Individual rules read very well on their own; a large unordered set is hard to see as a whole.

- **The link worth stating for the marks:** a decision tree can always be converted into a rule set by taking **one rule per root-to-leaf path**, ANDing the conditions along it. The reverse is *not* true, a general rule set need not correspond to any tree.
- So the real difference is **direct vs indirect induction**: rule-based classifiers learn rules straight from the data, decision trees learn a partition and get rules as a by-product.

3.4 Nearest Neighbor Classifier

What is a Nearest Neighbor Classifier? Main issues? Propose a fix. 1+3+4 3+6

PIN 4/23 81 Bh, 80 Ba, 80 Bh, 70 Asa 80 Bh is a numerical → companion.

Idea

- **Lazy learner** / instance-based: stores all training tuples, builds **no model**.
- All work happens at query time.
- “Tell me who your neighbours are and I will tell you who you are.”

Choosing k

- k too small → sensitive to **noise**, overfits.
- k too large → neighbourhood includes other classes, underfits.
- Pick by cross-validation; rule of thumb $k \approx \sqrt{n}$.

Algorithm (KNN)

1. Choose k and a distance measure (usually Euclidean).
 2. Compute the distance from the query tuple to **every** training tuple.
 3. Take the k smallest → the k nearest neighbours.
 4. **Classification**: assign the **majority class** among those k .
Prediction: take their **mean** value.
- **Normalise the attributes first**, else large-range attrs dominate the distance.
 - k odd for 2 classes → avoids ties.

Issues (80 Ba asks exactly this)

- **Expensive at query time:** $O(n)$ distance computations *per* query, no model to shortcut it.
- **Curse of dimensionality:** distances lose meaning in high dims.
- **Scale sensitive:** needs normalisation.
- **Irrelevant attributes** distort distance equally w/ relevant ones.
- Needs the whole training set in memory.

Fixes / classifier to propose

- Speed: k-d trees, ball trees, partial distance, editing (drop redundant tuples).
- **Attribute weighting** or feature selection for irrelevant attrs.
- **Distance-weighted voting:** nearer neighbours count more, $w = 1/d^2$.
- **Propose an eager learner** instead: a **decision tree** or **Naive Bayes** builds the model once, then classifies in near-constant time, and ignores irrelevant attrs.

How the distance metric is used in an instance-based classifier 10

70 Asa "What are the properties of a distance metric? How is distance metric used in instance based classifier?" **Two chapters in one question:** the properties are §2.4, the use is here. **The bridge between them is the answer.** verified/added

The four properties, restated §2.4

1. **Non-negativity:** $d(p, q) \geq 0$.
 2. **Identity:** $d(p, q) = 0$ iff $p = q$.
 3. **Symmetry:** $d(p, q) = d(q, p)$.
 4. **Triangle inequality:** $d(p, r) \leq d(p, q) + d(q, r)$.
- A function with all four is a **metric**. Euclidean, Manhattan and Minkowski qualify; **cosine similarity does not** (it is a similarity, not a distance).
 - **Where the metric enters KNN, step by step:** compute $d(\text{query}, \text{every training tuple}) \rightarrow \text{sort} \rightarrow \text{take the } k \text{ smallest} \rightarrow \text{majority vote (or distance-weighted vote, } w = 1/d^2, \text{ which uses the actual magnitudes and not just the ordering)}$.
 - **Choice of metric changes the answer:** Euclidean for continuous attributes, **Manhattan** when attributes are independent scales, **Hamming** for nominal attributes, **Jaccard** for asymmetric binary. **Normalise first** — an attribute measured in thousands otherwise dominates the sum (§2.4).

Why an instance-based classifier needs each one

- **The classifier is the distance function.** KNN stores the tuples and does no other modelling, so every decision it makes is a comparison of distances. A defective metric is a defective classifier.
- **Identity** $\rightarrow$ a training tuple identical to the query is at distance 0, so it is always among the k nearest. Without it a tuple could fail to match itself.
- **Symmetry** $\rightarrow$ " q is near p " and " p is near q " agree, so the neighbourhood is well defined however you sort.
- **Triangle inequality** $\rightarrow$ **this is the one that pays:** it lets **k-d trees** and **ball trees** discard an entire branch without measuring the tuples in it, turning an $O(n)$ scan into roughly $O(\log n)$. Drop it and every query must compare against the whole training set.
- **Non-negativity** $\rightarrow$ "nearest" means smallest, with a floor at 0.

3.5 Bayesian Classifier

Explain Naive Bayesian classification with a suitable example. 8 6 10 3+7

TOP 9/23 81 Ba, 79 Bh, 78 Bh, 76 Ash, 75 Ash, 72 Ch, 71 Ch, 71 Shr, 69 Ch Numerical in 79 Bh and 72 Ch → companion.

Bayes theorem

$$P(H|X) = \frac{P(X|H) P(H)}{P(X)}$$

- X = the tuple (evidence), H = hypothesis “ X belongs to class C ”.
- $P(H)$ = **prior**: probability of the class before seeing X .
- $P(H|X)$ = **posterior**: after seeing X . **This is what we want.**
- $P(X|H)$ = **likelihood**. $P(X)$ = **evidence**.
- 76 Ash asks prior vs posterior directly.

Naive Bayes

- Classify X to the class C_i w/ the largest $P(C_i|X)$.
- $P(X)$ is the **same for every class** → ignore it; just maximise $P(X|C_i)P(C_i)$.
- **“Naive” = class-conditional independence assumption**: given the class, the attributes are assumed independent, so

$$P(X|C_i) = \prod_{k=1}^n P(x_k|C_i)$$

- That turns 1 hard joint probability into n easy counts. **This is the whole trick.**

Algorithm steps (write these, then compute)

1. Compute the prior $P(C_i) = |C_{i,D}|/|D|$ for each class.
2. For each attribute value x_k in X , compute $P(x_k|C_i)$ from the counts.
 - Categorical: $P(x_k|C_i) = |\text{tuples in } C_i \text{ with } x_k|/|C_{i,D}|$.
 - Continuous: assume Gaussian, $g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.
3. Multiply: $P(X|C_i) = \prod_k P(x_k|C_i)$.
4. Compute $P(X|C_i)P(C_i)$ for every class.
5. **Predict the class w/ the maximum value.**

Laplacian correction 78 Bh short note 4

- **Problem**: if any $P(x_k|C_i) = 0$, the whole product collapses to **zero**, wiping out all the other evidence.
- **Fix**: add 1 to every count (and k to each denominator, k = no. of values):

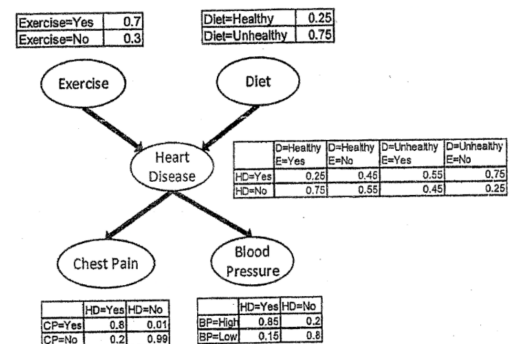
$$P(x_k|C_i) = \frac{n_k + 1}{|C_{i,D}| + k}$$

- Eg counts 0, 990, 10 → probabilities 0, 0.990, 0.010; after correction $\frac{1}{1003}, \frac{991}{1003}, \frac{11}{1003} = 0.001, 0.988, 0.011$.
- **Estimates barely move, but the zero is gone.**

Bayesian Belief Network (BBN) 3+5 10

81 Ba, 71 Ch 81 Ba is a numerical → companion.

- **Limitation of Naive Bayes**: the independence assumption is **almost never true** in practice. Correlated attributes get double-counted, which skews the posterior.
- **BBN drops it**: allows a *subset* of attributes to be conditionally dependent.
- Two parts:
 1. **Directed acyclic graph (DAG)**: node = variable, arc = direct causal dependence. Arc $Y \rightarrow Z$ means Y is a **parent** of Z .
 2. **Conditional probability table (CPT)** at each node, giving $P(Z | \text{parents}(Z))$.
- Joint probability: $P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(x_i))$
- A node is conditionally independent of its non-descendants given its parents.
- **When to use which** 71 Ch:
 - **Naive Bayes** → attributes roughly independent, need speed, little data, many attributes. Eg text/spam classification.
 - **BBN** → known causal structure and clear dependencies, need to reason w/ missing values. Eg medical diagnosis.



The 81 Ba network and its CPTs. Solved in the companion.

Advantages

- Fast to train and to classify; single pass over the data.
- Needs little training data.
- Handles missing values naturally (skip that factor).
- **Provably optimal** when the independence assumption actually holds.
- Incremental: counts can be updated as data arrives.

Disadvantages

- Independence assumption rarely holds → accuracy loss on correlated attrs.
- Zero-frequency problem (needs Laplacian correction).
- Cannot model interactions between attributes.
- Continuous attributes need a distribution assumption (usually Gaussian) or discretisation.

3.6 Artificial Neural Network Classifier

How does an ANN classifier work? Explain with algorithm. 8 2+6 5

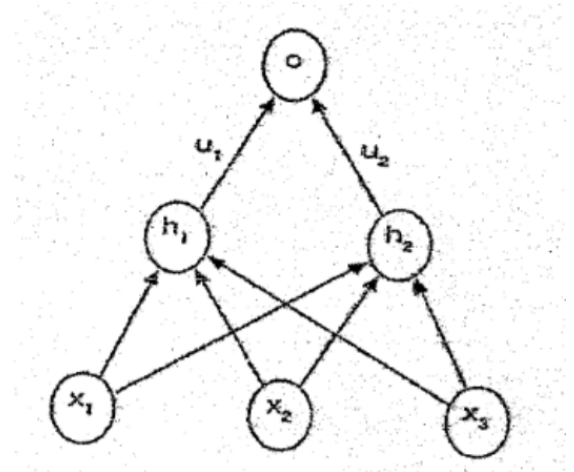
HOT 6/23 82 Bh, 81 Ba, 80 Ba, 76 Ch, 75 Ch, 72 Ka 76 Ch is a gradient numerical → companion.

Structure

- Set of connected **units (neurons)** in layers, each connection carrying a **weight**.
- **Input layer** (1 unit per attribute) → one or more **hidden layers** → **output layer** (1 unit per class).
- **Multilayer feed-forward**: no cycles, no connections within a layer.
- Each unit: net input = $\sum_i w_i x_i + \theta$ (bias), then an **activation function**.
- **Sigmoid** (logistic):

$$\sigma(\theta) = \frac{1}{1 + e^{-\theta}}, \quad \sigma'(\theta) = \sigma(\theta)(1 - \sigma(\theta))$$

- **That derivative identity is why sigmoid is used** → makes backprop cheap.



The 76 Ch network. Gradients derived in the companion.

Backpropagation algorithm

1. **Initialise** weights and biases to small random numbers.
2. **Forward pass**: compute net input and output for each unit, layer by layer.
3. **Compute error at the output**: $Err_j = O_j(1 - O_j)(T_j - O_j)$ where T_j = target.
4. **Backpropagate**: for a hidden unit, $Err_j = O_j(1 - O_j) \sum_k Err_k w_{jk}$.
5. **Update** weights and biases: $w_{ij} += l \cdot Err_j \cdot O_i$, $\theta_j += l \cdot Err_j$ (l = learning rate).
6. **Repeat** until the terminating condition (error small, or epoch limit).

General considerations 72 Ka asks this

- **Network topology**: no. of hidden layers and units. Chosen by trial and error; no rule.
- **Input encoding**: normalise inputs to $[0, 1]$; one unit per nominal value.
- **Output encoding**: 1 unit for 2 classes, else 1 per class.
- **Learning rate l** : too large → oscillation; too small → slow. Often $1/t$ w/ epoch t .
- **Initial weights**: small random, to break symmetry.
- **Update mode**: case updating (per tuple) vs epoch updating (per pass).

Advantages

- High tolerance of **noisy** data.
- Classifies patterns it was not trained on.
- Works when the relationship is unknown / non-linear.
- Well suited to continuous-valued inputs and outputs.
- Inherently **parallel**.

Disadvantages

- **Black box**: poor interpretability, hard to justify a decision.
- Long training times.
- Many parameters set empirically (topology, l , epochs).
- Needs numeric input → encoding effort.
- Can get stuck in a **local minimum**.

3.7 Issues: Overfitting, Validation, Model Comparison

Confusion matrix: calculate accuracy, sensitivity, specificity, precision, recall. 4 10 3+5

TOP 11/23 82 Bh, 80 Ba, 79 Ba, 78 Bh, 75 Ash, 74 Ash, 74 Ch, 73 Ch, 73 Shr, 72 Ch, 70 Ch **8 papers. Highest-yield numerical in the whole syllabus** → companion.

The 4 building blocks

- **TP**: positive tuples correctly labelled positive.
- **TN**: negative tuples correctly labelled negative.
- **FP**: negative tuples wrongly labelled positive (*false alarm*).
- **FN**: positive tuples wrongly labelled negative (*miss*).
- $P = TP + FN$ (all actual positives), $N = FP + TN$ (all actual negatives).

		Predicted class		Total
		yes	no	
Actual class	yes	TP	FN	P
	no	FP	TN	N
Total		P'	N'	P + N

All the formulas (one table, learn it whole)

Measure	Formula
Accuracy	$(TP + TN)/(P + N)$
Error rate	$(FP + FN)/(P + N) = 1 - \text{accuracy}$
Sensitivity, recall, TPR	TP/P
Specificity, TNR	TN/N
False alarm rate, FPR	$FP/N = 1 - \text{specificity}$
Precision	$TP/(TP + FP)$
F_1 score	$\frac{2 \times \text{prec} \times \text{rec}}{\text{prec} + \text{rec}}$
F_β	$\frac{(1 + \beta^2) \text{prec} \times \text{rec}}{\beta^2 \text{prec} + \text{rec}}$

- **Recall = sensitivity = TPR**. Three names, one thing. Papers use all three.
- F_1 = harmonic mean of precision and recall.

When accuracy is the wrong measure 80 Ba 3

- **Class imbalance**. If 99% of tuples are negative, a classifier that always says “negative” scores **99% accuracy** and is useless.
- Examples: cancer screening, fraud detection, intrusion detection, defect detection.
- **Use instead**: sensitivity + specificity separately, precision/recall, F_1 , ROC/AUC, or cost-sensitive measures.
- Note: $\text{accuracy} = \text{sensitivity} \frac{P}{P+N} + \text{specificity} \frac{N}{P+N} \rightarrow$ accuracy hides the split.

THE trap in this chapter. Han & Kamber put **actual** on the rows, **predicted** on the columns. The IOE papers **flip this between years**: 74 Ch and 75 Ash write “Actual \ Predicted”, but 78 Bh and 72 Ch write “Predicted / Actual”.

If you read the orientation wrong, **FP and FN swap**, so sensitivity and specificity swap and precision is wrong. **Always copy the row/column labels onto your answer sheet first**, mark which axis is actual, then compute.

verified/added

Precision vs recall trade-off 74 Ch asks the inverse relation

- Loosen the threshold → more positives flagged → recall ↑, precision ↓.
- Tighten it → precision ↑, recall ↓.
- Perfect precision *and* recall is only possible w/ a perfect classifier → report F_1 to balance them.

ROC curve 73 Shr 1

- Plot of **TPR (y) vs FPR (x)** as the decision threshold sweeps.
- Diagonal = random guessing. Top-left corner = perfect.
- **AUC** = area under the curve. Closer to 1 = better; 0.5 = useless.
- Lets you compare 2 classifiers **independently of the threshold** and of class balance.

Evaluating a classifier: holdout, random subsampling, k -fold, bootstrap 8

79 Ba “Describe the working mechanism as well as the merits and demerits of the holdout method, random sampling, k -cross validation and bootstrap approaches in evaluating the performance of a classifier.” **8 marks means all four, each with mechanism + merit + demerit.** verified/added

Method	Working mechanism	Merits	Demerits
Holdout	Partition D once at random into a training set and a disjoint test set, typically $\frac{2}{3}$ / $\frac{1}{3}$. Build the model on the first, measure accuracy on the second.	Simplest and cheapest : one model is built. Test set is genuinely unseen.	Estimate depends entirely on one lucky or unlucky split (high variance). Only part of the data trains, so accuracy is pessimistic on small D .
Random subsampling	Holdout repeated k times with a fresh random split each time; report the mean of the k accuracies.	Averages the split luck away $\rightarrow$ much more stable than one holdout. Simple to run.	k models to build. Splits overlap uncontrolled, so some tuples are tested many times and others never , and the runs are not independent.
k-fold cross-validation	Split D into k equal, mutually exclusive folds. In round i train on all folds but D_i and test on D_i . Accuracy = total correct over all k rounds / $ D $. $k = 10$ is the recommended value ; stratified folds keep each class's proportion.	Every tuple is tested exactly once and trains $k - 1$ times $\rightarrow$ uses the data fully and has low bias and low variance. The standard method, and the safe answer to “which would you use”.	Costs k model builds. Still slightly pessimistic (trains on $\frac{k-1}{k}$ of D). Leave-one-out ($k = n$) removes the bias but is prohibitively expensive and has high variance.
Bootstrap (.632)	Sample n tuples from D with replacement to form the training set; the tuples never drawn become the test set. A tuple is missed with probability $(1 - \frac{1}{n})^n \approx e^{-1} = 0.368$, so 63.2% of D trains on average. Repeat b times and combine: $acc = \frac{1}{b} \sum_{i=1}^b (0.632 acc_{test_i} + 0.368 acc_{train_i})$.	Best estimator for small data sets , where holdout wastes too much and folds are tiny. Reuses every tuple.	Optimistically biased , because duplicated training tuples also reappear in the model. The 0.632 weighting is a correction, not a cure. Most expensive of the four.

- **One-line ranking to close the answer**: holdout for speed, cross-validation for a trustworthy number, bootstrap when D is small.

Improving accuracy: ensembles (1 line each is enough)

- **Bagging**: train k models on bootstrap samples, take a **majority vote**. Reduces *variance*; helps unstable learners like decision trees.
- **Boosting / AdaBoost**: train models **sequentially**, each re-weighting the tuples the previous one got wrong; weighted vote. Reduces *bias*; can overfit noise.
- **Random forest**: bagging + a random attribute subset at each split. Robust to overfitting, handles many attributes.

3.8 PYQ Mapping

N marks a question fully solved in the Ch 3 Numerical companion.

3.1 Basics

1. **PIN 3/23** Block diagram of classification stages + inverse precision/recall relation + confusion matrix **N**. 2+3+5 74 Ch
2. Why is classification supervised? + impurity measures. 10 71 Shr
3. When do we use a classifier? (asked w/ the KNN numerical) 3 80 Bh

3.2 Decision Tree

4. **N** Draw a decision tree using **ID3**. 10 79 Bh
5. **N** Construct a tree using **information gain** + predict a new tuple. 8 78 Bh
6. **N** Which attribute would ID-3 choose as root? + draw the full tree. 2+5 76 Ch
7. **TOP /23** Decision tree + how information gain selects attributes, w/ example. 8 73 Shr
8. Accuracy of classifiers + how to select the best root attribute. 4+6 76 Ash
9. **PIN 2/23** Gini-Index w/ example + handling overfitting. 6+4 75 Ch 10 71 Ch
10. **PIN 1/23** **Hunt's Algorithm** + test conditions per attribute type. 10 72 Ka
11. **PIN 1/23** Decision tree w/ the concept of Naive Bayes classification. 10 74 Ch

3.3 Rule Based

12. **HOT 4/23** How a rule-based classifier works, w/ example. 7 80 Bh 2+8 74 Ash
13. **PIN 1/23** Rule-based classifier + **CN2** + define **Accuracy** and **Laplace** measures. 9 70 Ch
14. **PIN 1/23** Why is conflict resolution necessary? + common strategies. 2+4 76 Ch

3.4 Nearest Neighbor

15. **N** Classify a flower w/ **KNN**, $K = 3$, Euclidean. 3+6 80 Bh
16. **PIN 1/23** What is KNN? Main issues? Propose a classifier that solves them. 1+3+4 80 Ba

3.5 Bayesian

17. **TOP /23** Naive Bayes w/ suitable example. 8 75 Ash 6 78 Bh
18. **N** Classify $X = (\text{Home Owner}=\text{No}, \text{Marital Status}=\text{Married}, \text{Income}=\$120\text{K})$. 6 79 Bh
19. **N** Predict class for $X = (\text{age}=\text{youth}, \text{income}=\text{medium}, \text{student}=\text{yes}, \text{credit}=\text{fair})$. 10 72 Ch
20. **PIN 1/23** Prior and posterior probabilities + algorithmic steps + strengths. 3+7 76 Ash
21. **PIN 1/23** Bayes classifier + when to use Naive Bayes vs BBN. 10 71 Ch
22. **PIN 1/23** Naive Bayes + solving overfitting in classification. 10 71 Shr
23. **N** Limitation of Naive Bayes + how BBN overcomes it + heart-disease inference. 3+5 81 Ba
24. **PIN 1/23** Short note: **Laplacian correction**. 4 78 Bh

3.6 ANN

25. **HOT 3/23** How does an ANN classifier work? w/ algorithm. 8 81 Ba, 75 Ch 2+6 72 Ka
26. **N** Compute activation at h_1 , $\partial E / \partial o$, $\partial E / \partial w_{12}$. 2+3+4 76 Ch
27. **PIN 1/23** Short note: Neural Network Classifier. 5 80 Ba

3.7 Issues: Overfitting, Validation, Model Comparison

28. **N** **TOP 7/23** **Confusion matrix numericals** (8 papers, same method, different tables):
 - Classification error, sensitivity, false alarm rate, specificity 3+5 80 Ba
 - Accuracy, TPR, FPR, precision 4 78 Bh
 - Accuracy, sensitivity, specificity, precision 10 75 Ash, 74 Ash
 - + recall 10 72 Ch
 - ROC + TPR, FPR, precision 1+3+6 73 Shr
 - Build the matrix from 2 predicted sequences, then 4 metrics 10 70 Ch
 - Given matrix $\rightarrow$ accuracy, sensitivity, precision 5 74 Ch
29. **PIN 1/23** When can you not use accuracy? Give examples. 3 80 Ba
30. **PIN 1/23** Short note: Overfitting problem in classification. 4 78 Bh

Papers added by the 2026-09 merge

The six papers of [DM4.1BCT.pdf](#) (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. [verified/added](#)

- **N** When do we use a decision tree? + a full **ID-3** tree on the credit-risk table + how to evaluate the model. 2+9+3 82 Bh
- **N** Decision-tree algorithm + use **ID3** to identify the root node only. 2+8 81 Bh
- Working mechanism *and* merits and demerits of holdout, random sampling, k -fold cross-validation and bootstrap. 8 79 Ba [verified/added](#)
- **N** Confusion matrix $\rightarrow$ accuracy, error rate, sensitivity, specificity, precision, recall. 4 79 Ba
- **N** What is ID3? + TPR, FPR and accuracy from a matrix that **labels both axes Predicted**. 2+6 73 Ch

- Importance of the **homogeneity measure** in a decision tree + explain **Gini index**. 8 70 Asa
- How is the distance metric **used in an instance-based classifier**? 10 70 Asa verified/added
- How is a decision tree classifier **different from a rule-based classifier**? 8 69 Ch verified/added
- Bayes' theorem + how Naive Bayes **simplifies the computational complexity**. 12 69 Ch
- Nearest neighbour classifier and when it is used 81 Bh · short notes Neural Network Classifier 82 Bh, Overfitting and ROC
73 Ch. 2+4 5 4

Coverage check

- All Ch 3 PYQs covered. **12 are numerical** → all solved in the companion.
- **Gaps filled by research** verified/added :
 - **Laplace and m-estimate** rule measures and the **CN2** algorithm 70 Ch: absent from every source note; taken from Tan/Steinbach and Han & Kamber §8.4.
 - **Conflict resolution strategies** 76 Ch: only named in the notes, not explained.
 - The **confusion-matrix orientation trap** (papers flip actual/predicted between years).

Sources used for this chapter

- **Han & Kamber 3rd ed**, ch 8 (basic concepts, decision tree induction, attribute selection measures, Bayes, rule-based, model evaluation) and ch 9 (BBN, backpropagation, lazy learners). **Primary authority; every formula above checked against it.**
- **DBK Lectures 5–11** (Thapathali): Hunt's algorithm, tree construction, Naive Bayes, KNN, ANN, evaluation.
- `data mining_/Chapter 3 - Classification.pdf`, `hunts algorithm.pdf`, `ClassificationDecisionTree.ppt`, `Chapter8_2.ppt`, `Bayes_Net.docx`.

4. Association Analysis

10 hrs · in 23/23 papers · 5 topics, 3 TOP tier · **heaviest numericals in the syllabus** → Ch 4 companion.

Weight of the chapter:

- FP-Growth / FP-Tree **TOP 17/23** → **2nd most-asked topic in the whole subject.**
- Frequent Itemset & Apriori **TOP 14/23** → almost always a full numerical.
- Basics (support, confidence, market basket) **TOP 10/23**
- Sequential / Subgraph / Infrequent patterns **HOT 5/23** · Categorical attributes **PIN 3/23**

4.1 Basics and Algorithms

What is association analysis? Explain its importance in market-basket analysis. **2+5 10 2+6**

TOP 10/23 82 Bh, 80 Ba, 75 Ch, 75 Ash, 74 Ch, 73 Ch, 72 Ch, 72 Ka, 70 Ch, 70 Asa

Definitions

- **Itemset**: a set of 1 or more items. k -itemset = contains k items.
- **Transaction T** : a set of items bought together. D = the whole database.
- **Association rule**: an implication $A \Rightarrow B$ where $A \subset I$, $B \subset I$, and $A \cap B = \emptyset$.
- **Association analysis**: finding all rules that satisfy a minimum support and a minimum confidence.

The 2 measures (learn both definitions exactly)

$\sigma(X)$ = support count of X (how many transactions contain it), $|D|$ = total transactions.

$$\text{sup}(A \Rightarrow B) = P(A \cup B) = \frac{\sigma(A \cup B)}{|D|}$$

$$\text{conf}(A \Rightarrow B) = P(B|A) = \frac{\sigma(A \cup B)}{\sigma(A)}$$

- **Support** → how *frequent*. Filters out rules with too little evidence.
- **Confidence** → how *reliable*. Strength of the implication.
- A rule is **strong** if it meets *both* thresholds.
- **Support is symmetric** ($A \cup B$), **confidence is not** ($A \Rightarrow B \neq B \Rightarrow A$).

Market basket analysis 75 Ch short note

- Analyses which items customers put in the basket *together*.
- Classic result: {nappies} $\Rightarrow$ {beer}.
- **Uses**: shelf layout, cross-selling, product bundling, catalogue design, promotion planning, recommendation ("customers who bought this also bought...").
- **Actionable both ways**: put them together to increase convenience, or far apart to force the customer past other shelves.

2-step process (every algorithm does this)

1. **Find all frequent itemsets**: those with support $\geq \text{min_sup}$. **This is the expensive step** and is what Apriori / FP-Growth optimise.
2. **Generate strong rules** from them: for each frequent itemset l , take every non-empty proper subset s and output $s \Rightarrow (l - s)$ if $\frac{\text{sup_count}(l)}{\text{sup_count}(s)} \geq \text{min_conf}$.
3. A k -itemset yields $2^k - 2$ candidate rules.

Applications beyond retail 80 Ba asks where it is applicable

- Web usage / click-stream mining, medical symptom-disease co-occurrence, **gene co-occurrence** (Ch 1), fraud detection, intrusion detection, text mining.

Pattern evaluation: why support-confidence is not enough 8

70 Ch “Why is pattern evaluation important? Explain the statistical based measures used for measuring interestingness.” **Full worked example in the companion.**

The problem

- A rule can clear both thresholds and still be **completely misleading**.
- **Classic case:** 10,000 txns, 6,000 contain *game*, 7,500 contain *video*, 4,000 contain both.
 - $support = 4000/10000 = 40\%$ (passes)
 - $confidence(game \Rightarrow video) = 4000/6000 = 66\%$ (passes)
 - But $P(video) = 75\%$ on its own, **higher than 66%**.
 - Buying a game makes you *less* likely to buy a video. The “strong” rule is **negatively correlated**.
- Cause: **confidence ignores the support of the consequent**.

Correlation measures verified/added

$$lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{conf(A \Rightarrow B)}{sup(B)}$$

- > 1 positively correlated · < 1 negative · $= 1$ independent.
- Above: $lift = 0.40/(0.60 \times 0.75) = 0.89 < 1 \rightarrow$ caught.

$$\chi^2 = \sum \frac{(observed - expected)^2}{expected}$$

- Above: $\chi^2 = 555.6$, and observed $<$ expected $\rightarrow$ negative correlation.

4 null-invariant measures

- $all_conf(A, B) = \frac{sup(A \cup B)}{\max\{sup(A), sup(B)\}} = \min\{P(A|B), P(B|A)\}$
- $max_conf(A, B) = \max\{P(A|B), P(B|A)\}$
- $Kulc(A, B) = \frac{1}{2}(P(A|B) + P(B|A))$
- $cosine(A, B) = \frac{sup(A \cup B)}{\sqrt{sup(A) \times sup(B)}}$
- **Null-invariant** = unaffected by the number of transactions containing *neither* item. **lift** and χ^2 are **NOT null-invariant**, so they distort on sparse data.
- Use **Kulc** + **imbalance ratio** together for the clearest reading.

Why a 100 % confidence rule is uninteresting, and why 99 % is 4

80 Ba “Generally we are more interested in rules with high confidence. However often we will not be interested in rules with a confidence of 100 %. Why? Then specifically explain why rules with 99 % confidence may be interesting.” verified/added

100 %: true by construction, so it teaches nothing

- $conf(A \Rightarrow B) = 100\%$ means **every** transaction holding *A* also holds *B*, no exceptions anywhere in *D*.
- Almost always that is a **property of the schema, not of the customers** $\rightarrow$ already known before mining started:
 - **Definitional:** $\{pregnant\} \Rightarrow \{female\}$, $\{Corolla\} \Rightarrow \{car\}$.
 - **Taxonomic:** $\{Kathmandu\} \Rightarrow \{Nepal\}$, item $\rightarrow$ its own category.
 - **Subset artefact:** $\{milk, bread\} \Rightarrow \{milk\}$. Consequent already sits inside the antecedent’s itemset.
 - **Bundled by a business rule:** free gift always shipped with the product, so the pair can never come apart.
- **No headroom, so no action.** You cannot lift sales of *B* by promoting *A*: everyone who takes *A* takes *B* already. Cross-selling, shelf layout and recommendation all have nothing to work with.
- Such rules are also generated in bulk (every frequent itemset spawns its own subset rules) and **bury the genuine findings**. Fix: a maximum support threshold, or drop rules whose consequent is implied by the antecedent.

99 %: real regularity, and the 1 % is the payload

- A definitional rule cannot have exceptions. So **99 % proves the dependency is empirical**, discovered from behaviour rather than built into the data dictionary.
- It is still near-deterministic $\rightarrow$ strong enough to predict on, unlike a 60 % rule.
- **The interest is in the 1 % that breaks it**, and it reads two ways. Give both:
 1. **The exceptions are wrong** $\rightarrow$ candidate **data-quality defects**: mis-keyed records, typos, a faulty sensor, a duplicated ID. Near-100 % rules are used exactly this way to discover integrity constraints and clean a warehouse (Ch 2).
 2. **The exceptions are real** $\rightarrow$ candidate **anomalies**: fraud, intrusion, a defective batch, a customer segment behaving unlike every other. This is the association-rule route into anomaly detection (Ch 6).
- Practical value too: 1 % is small but non-empty, so the customers who take *A* without *B* are a **precisely targetable list**.
- **One line worth memorising:** a 100 % rule tells you about the *schema*, a 99 % rule tells you about the *world*, and its exceptions are where the interesting cases live.

4.2 Frequent Itemset Pattern & Apriori Principle

Explain the Apriori algorithm / principle with example. 2+6 10 8

TOP 14/23 82 Bh, 81 Bh, 80 Bh, 80 Ba, 79 Bh, 79 Ba, 76 Ch, 76 Ash, 74 Ash, 73 Ch, 71 Ch, 71 Shr, 70 Ch, 69 Ch Numerical in 7 of those → companion.

The Apriori property (state this, it is the answer)

- All non-empty subsets of a frequent itemset must also be frequent.
- Contrapositive, and the one actually used: **if an itemset is infrequent, all of its supersets are infrequent** → prune them without counting.
- This is **anti-monotonicity**: support never increases as an itemset grows.
- **Effect**: turns a 2^n brute-force search into something tractable. 71 Ch asks exactly this, 10.

Brute force, for contrast

- n items → $2^n - 1$ candidate itemsets, each needing a database scan.
- Apriori prunes the lattice level by level instead.

Advantages

- Simple, easy to implement and to explain.
- Uses a level-wise search that is easy to parallelise.
- The pruning is **provably safe** (no frequent itemset is ever lost).

Algorithm

1. Scan D , count each item → keep those $\geq \text{min_sup}$ → L_1 .
2. Repeat for $k = 2, 3, \dots$ until L_{k-1} is empty:
 - (a) **Join**: $C_k = L_{k-1} \bowtie L_{k-1}$. Join 2 itemsets only if their **first $k - 2$ items are identical** (keeps items sorted, avoids duplicates).
 - (b) **Prune**: drop any candidate having any $(k - 1)$ -subset not in L_{k-1} .
 - (c) **Count**: scan D once, count the survivors.
 - (d) $L_k =$ candidates meeting min_sup .
3. Answer = $\bigcup_k L_k$.

Then generate rules

- For each frequent l , for each non-empty proper subset s : output $s \Rightarrow (l - s)$ if $\frac{\text{sup}(l)}{\text{sup}(s)} \geq \text{min_conf}$.
- **When no rule can exist at any confidence** 79 Ba 3
 verified/added : a rule needs a *proper* non-empty subset, so it needs a frequent itemset of size ≥ 2 . If $L_2 = \emptyset$ (every frequent itemset is a single item), then **no rule exists for any confidence threshold, including 0 %** → lowering min_conf cannot help, only lowering min_sup can. Check L_2 before answering, never the confidence value.

Disadvantages (the setup for FP-Growth)

- **Generates a huge number of candidates**. 10^4 frequent 1-itemsets → about 10^7 candidate 2-itemsets.
- **Requires repeated database scans** — one per level, so $k + 1$ scans for a k -itemset. Very costly on disk-resident data.
- Struggles with long patterns and with low min_sup .

4.3 FP-Growth and the FP-Tree

What is FP-Growth? Explain with example. How does it overcome Apriori's problems? 2+5 8 10 2+6

TOP 17/23 82 Bh, 81 Bh, 81 Ba, 80 Bh, 79 Bh, 79 Ba, 78 Bh, 75 Ch, 75 Ash, 74 Ch, 74 Ash, 73 Ch, 73 Shr, 72 Ka, 71 Shr, 70 Asa, 69 Ch **The single most-asked topic in this chapter.**

Key idea

- **Generate NO candidates at all.**
- Compress D into a **frequent-pattern tree (FP-tree)**, then mine that tree by **divide and conquer**.
- Only **2 database scans** in total, against Apriori's $k + 1$.

Step 1: build the FP-tree

- Scan 1:** count every item. Discard the infrequent ones.
- Sort the frequent items by **descending support** → the **F-list**. **Break ties consistently** (alphabetically) and say so.
- Scan 2:** for each transaction, keep only frequent items, **reorder them into F-list order**, and insert that path into the tree.
 - Share a prefix with an existing path → just increment those counts.
 - Otherwise create new nodes with count 1.
- Maintain a **header table**: one entry per frequent item, with a **node-link** chain threading every occurrence of that item in the tree.
 - **Why descending order:** the most common items sit near the root, so paths share prefixes and the tree stays small.

Step 2: mine the tree (FP-Growth)

- Work **bottom-up** through the header table, ie *least* frequent item first.
- For item α , follow its node-links to collect its **conditional pattern base** = every prefix path leading to α , each carrying α 's count at that node.
- Build the **conditional FP-tree** from that base, keeping only items that are still frequent *within* it.
- Recurse** on the conditional tree, concatenating α to each pattern found.
- If a conditional tree is a **single path**, just enumerate all subsets of that path.

Apriori vs FP-Growth (asked as a direct comparison 4 times)

	Apriori	FP-Growth
Candidates	Generates them	None
DB scans	$k + 1$	2
Strategy	Breadth-first, level-wise	Depth-first, divide & conquer
Structure	Hash tree	FP-tree (prefix tree)
Memory	Low	Tree must fit in memory
Speed	Slower	Roughly an order faster

- **FP-Growth's weakness:** if the tree does not fit in memory, or the data is very sparse so paths barely share prefixes, the advantage disappears.

4.4 Handling Categorical Attributes

How do you handle categorical attributes in data mining? Issues and fixes. 3+3 2+3+3 4

PIN 3/23 79 Bh, 73 Shr, 72 Ch

The problem

- Association analysis assumes **asymmetric binary** items (present / absent).
- Real tables hold categorical (Gender, Region) and continuous (Age, Income) attributes.

Handling: binarisation

- Convert each (attribute, value) pair into one binary item.
- Eg Gender → items Gender=Male, Gender=Female.
- Continuous attributes must be **discretised** first → Ch 2, §2.2, then binarised.

Issues that arise (this is what 73 Shr wants)

- **Too many items:** an attribute with many values explodes the item count. **Fix:** group rare values into an “Other” category.
- **Many items are infrequent:** rare values never reach min_sup, so patterns involving them are lost. **Fix:** lower min_sup, or use a per-item threshold.
- **Some items are far too frequent:** eg Gender=Male at ≈50% produces masses of trivial redundant rules. **Fix:** set a maximum support threshold too.
- **Redundancy:** values of one attribute are **mutually exclusive**, so an itemset containing Gender=Male and Gender=Female has support 0 by construction. **Fix:** never generate candidates that pair 2 values of the same attribute.
- **Loss of ordering** for ordinal attributes after binarisation.

4.5 Sequential, Subgraph and Infrequent Patterns

Sequential patterns, subsequences, subgraph patterns 3+3 2 3

HOT 5/23 79 Bh, 76 Ch, 76 Ash, 73 Ch, 72 Ka Subsequence and subgraph enumeration are numericals → companion.

Sequential patterns

- A **sequence** is an *ordered* list of **elements**; each element is an unordered *set* of items. $s = \langle e_1 e_2 \dots e_n \rangle$.
- **Length** = number of *items*; a *k*-sequence has *k* items. **Not the number of elements** — this is the usual mistake.
- **Subsequence:** $\langle a_1 \dots a_m \rangle$ is a subsequence of $\langle b_1 \dots b_n \rangle$ if there exist $i_1 < i_2 < \dots < i_m$ with $a_1 \subseteq b_{i_1}, \dots, a_m \subseteq b_{i_m}$.
 - **Order across elements must be preserved; within an element it need not be.**
- **Support** of a sequence = fraction of data sequences containing it.
- **Algorithms:** AprioriAll 76 Ash short note, GSP, SPADE, PrefixSpan. AprioriAll applies the same anti-monotone pruning to sequences.
- **Uses:** web click-streams, purchase sequences over time, DNA, telephone call patterns.

Subgraph patterns 79 Bh 2

- Data = a collection of **graphs** (molecules, web fragments, social networks).
- A **frequent subgraph** is one appearing in at least min_sup of them.
- **Candidate generation by growing**, 2 schemes:
 - **Vertex growing:** join 2 graphs sharing a common subgraph, add 1 vertex.
 - **Edge growing:** join 2 graphs sharing a core, add 1 edge. **Preferred** — it generates fewer candidates.
- **The hard part is graph isomorphism:** deciding whether 2 drawings are the same graph. Handled with a canonical labelling.
- Algorithms: AGM, FSG, gSpan.

Infrequent / negative patterns

- A **negative pattern** is a set of items that appear together *far less* often than independence would predict.
- Eg {Coke, Pepsi} — customers buy one or the other.
- Useful for detecting substitute products and for fraud (a combination that should never occur).
- **Hard because** support is anti-monotone, so “infrequent” cannot be pruned the same way.

4.6 PYQ Mapping

N = solved in the Ch 4 Numerical companion.

4.1 Basics

1. TOP 7/23 Association analysis + importance in market-basket analysis. 2+5 72 Ka 10 72 Ch 2+6 80 Ba
2. PIN 2/23 Why is association analysis required? + Apriori principle w/ example. 2+6 75 Ash, 74 Ch
3. PIN 1/23 Importance of SUPPORT and CONFIDENCE + FP-Growth w/ example. 10 72 Ch
4. N PIN 1/23 Why is pattern evaluation important? + statistical interestingness measures. 8 70 Ch
5. PIN 1/23 Why 100% confidence is uninteresting but 99% is. 4 80 Ba
6. PIN 1/23 Short note: Market Basket Analysis. 4 75 Ch

4.2 Apriori

7. N Apriori, min support count 4, confidence 80%, 7 transactions. 2+7 80 Bh
8. N Candidate + large itemsets, min support 2. 4+5 80 Ba
9. N Derive association rules, min sup 50%, min conf 80%. 8 79 Bh
10. N Fast-food orders, min sup 2/9, min conf 7/9; frequent k -itemsets + strong rules. 5+3 76 Ch
11. N L-itemsets at $L = 3$ w/ Apriori + F-list w/ FP-Growth. 12 76 Ash
12. N Candidate, frequent itemsets and rules, min sup 20%, min conf 80%. 8 74 Ash
13. N Apriori w/ support 50%. 10 70 Ch
14. PIN 1/23 How does Apriori optimise the brute-force approach? 10 71 Ch
15. PIN 1/23 What are association rules? How does Apriori generate them? 10 71 Shr

4.3 FP-Growth

16. TOP 7/23 What is FP-Growth? Explain w/ example. 2+5 80 Bh 8 74 Ash 10 71 Shr 2+6 75 Ash 1+8 72 Ka 2+5 81 Ba
17. N Draw FP-tree for the given 5-transaction table. 4+4 75 Ch
18. N Limitation of Apriori vs FP-Growth + $\{M, O, N, K, E, Y\}$ database, min sup 60%, min conf 80%. 2+7 81 Ba
19. N Build FP-tree transaction by transaction + mine it, threshold 33.34% / 60%. 4+4 78 Bh
20. N How does FP-Growth overcome Apriori? + generate the FP-tree. 2+8 74 Ch
21. N FP-tree + association rules, support 30%, confidence 75%. 10 73 Shr
22. PIN 1/23 Short note: FP-Tree. 5 79 Bh

4.4 / 4.5 Categorical, Sequential, Subgraph

23. N Handle categorical attributes + generate 4 subsequences from $\{\{2, 3, 5\}, \{6, 7, 8\}, \{9, 1\}, \{7, 4\}\}$. 3+3 79 Bh
24. PIN 1/23 Categorical data, issues, how to handle them. 2+3+3 73 Shr
25. N List all 4-subsequences of $\{\{1, 3\}\{2\}\{2, 3\}\{4\}\}$ + draw candidate subgraphs by edge growing. 3+3 76 Ch
26. PIN 1/23 What are subgraph patterns? 2 79 Bh
27. PIN 3/23 Short notes: AprioriAll 76 Ash · Sequential pattern 72 Ka · Categorical data 72 Ch. 3 4

Papers added by the 2026-09 merge

The six papers of DM4.1BCT.pdf (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. N = solved in this chapter's companion. verified/added

- N Support and confidence + derive rules, min sup 60%, min conf 80%, 5 transactions. 2+8 82 Bh
- N Apriori with **confidence 65%** and min support 50%, 4 transactions. 2+6 81 Bh
- N Apriori on a 7×9 **binary** table (min_sup 20%) + **can any rule exist at any confidence threshold?** 5+3 79 Ba
- verified/added
- N Apriori in market-basket analysis, both thresholds 50%. 3+7 73 Ch
- Advantages of FP-Growth over Apriori + FP-Growth w/ example. 2+6 81 Bh
- How does FP-Growth generate frequent itemsets **without candidate generation?** 6 79 Ba
- Use of FP-Growth in market-basket analysis 73 Ch · FP-Growth in detail 70 Asa. 10 12
- What is frequent itemset mining? How do Apriori *and* FP-Growth optimise the brute-force approach? 15 69 Ch
- How can Apriori be used to find rules *out of* a frequent itemset? 7 69 Ch
- Association rules + their importance w/ example 70 Asa · short note **FP Growth** 82 Bh. 6 5
- Sequential pattern and sub-graph pattern w/ suitable example. 4+4 73 Ch

Coverage check

- All Ch 4 PYQs covered. **14 are numerical** → companion.

- **Gaps filled by research** verified/added :
 - The **null-invariant measures** (all_conf, max_conf, Kulc, cosine) and the game/video counter-example for 70 Ch: your notes define only support and confidence.
 - **Issues with categorical attributes** 73 Shr: the mutual-exclusion and too-frequent-item problems are in no source note.
 - **Edge vs vertex growing** for subgraphs.

Sources used for this chapter

- **Han & Kamber 3rd ed** ch 6 (frequent patterns, Apriori, FP-Growth, pattern evaluation) and ch 7 (advanced pattern mining).
Primary authority.
- **DBK Lectures 12–14**, [data mining_/Chapter-4 Association Analysis.pdf](#), [Associations_Rule_mining.ppt](#), [FP-Growth algorithm.ppt](#), [FP Tree.pdf](#), [Episode_Mining.PDF](#).

5. Cluster Analysis

9 hrs · in 23/23 papers · 5 topics, 3 TOP tier · numericals → Ch 5 companion.

Weight of the chapter:

- Basics & types of clustering **TOP 10/23** · K-means **TOP 8/23** · DBSCAN **TOP 8/23**
- Hierarchical **HOT 4/23** · Evaluation / issues **HOT 4/23**
- **K-means numerical appears in 8 papers, hierarchical dendrogram in 4.**

5.1 Basics and Algorithms

What is cluster analysis? Applications? How does it differ from classification? **8 5 2+6 3+4**

TOP 13/23 81 Ba, 80 Bh, 79 Bh, 75 Ch, 74 Ash, 73 Ch, 73 Shr, 72 Ch, 72 Ka, 71 Ch, 71 Shr, 70 Asa, 69 Ch

Definition

- **Cluster** = a collection of objects **similar within** the group and **dissimilar to** objects in other groups.
- **Cluster analysis** = grouping objects so that intra-cluster similarity is **maximised** and inter-cluster similarity **minimised**.
- **Unsupervised**: no class labels are given. The classes are the *output*, not the input.
- Also called **unsupervised classification** or segmentation.

Clustering vs classification 72 Ka **5**

	Classification	Clustering
Learning	Supervised	Unsupervised
Labels	Given in training	None; discovered
Goal	Predict a known label	Find the groups themselves
Evaluation	Accuracy vs truth	Cohesion / separation
Classes	Fixed, known	Number often unknown

Applications asked in 5 papers

- **Market segmentation**: group customers by behaviour.
- **Document / search-result grouping**, topic discovery.
- **Image segmentation** and pattern recognition.
- **Biology**: taxonomy of species, gene expression grouping.
- **Land use** from satellite data; city planning; earthquake epicentre grouping.
- **Outlier detection** — points in no cluster → Ch 6.
- **Data reduction / preprocessing** step for other algorithms.

Requirements of a good clustering method

- Scalability; handles **mixed attribute types**; finds **arbitrary shapes**.
- Minimal domain knowledge for parameters; copes with **noise and outliers**.
- **Insensitive to input order**; handles high dimensionality; interpretable and usable.

Types of clusters (81 Ba asks this directly)

- **Well-separated**: every point is closer to all points in its own cluster than to any point outside. The ideal case.
- **Prototype-based (centre-based)**: each point is closer to its cluster's **centroid** (mean) or **medoid** (most central actual point) than to any other. → what K-means finds.
- **Contiguity-based (nearest neighbour)** ^{71 Shr 10}: each point is closer to at least **one point** in its cluster than to any point outside. Handles chains and irregular shapes; **breaks down when noise bridges 2 clusters**. → single-link hierarchical, or DBSCAN.

- **Density-based**: a cluster is a **dense region** separated by low-density regions. Handles arbitrary shapes and noise. → DBSCAN.
- **Conceptual (shared-property)**: points share some general property, not just proximity. Needs a concept definition.

Types of clustering methods

- **Partitioning**: K-means, K-medoids/PAM, CLARA.
- **Hierarchical**: AGNES, DIANA, BIRCH, Chameleon.
- **Density-based**: DBSCAN, OPTICS, DENCLUE.
- **Grid-based**: STING, CLIQUE, WaveCluster.
- **Model-based**: EM, COBWEB, SOM.

Grid-based, in 3 steps (NEW)

- **Partition** the data space into a multi-dim grid of finite cells.
- **Compute cell statistics**: count, mean, density per cell.
- **Merge** adjacent cells above a density threshold into clusters.
- **Why it is the fast one**: cost depends on the **number of cells**, not the number of points → scales to huge datasets. Price paid: resolution is capped by cell size.

5.2 K-means Clustering

Write the K-means algorithm. Explain with example and state its limitations. 2+6 2+8 4+6 4+4

TOP 13/23 82 Bh, 81 Bh, 81 Ba, 80 Ba, 79 Ba, 78 Bh, 76 Ch, 75 Ash, 74 Ch, 74 Ash, 73 Ch, 71 Shr, 69 Ch **Numerical in 7 of those** → companion.

Algorithm (write these 5 steps, then compute)

1. Choose K , the number of clusters.
2. **Initialise** K centroids (usually K randomly chosen data points).
3. **Assign** each point to the cluster whose centroid is nearest (Euclidean).
4. **Recompute** each centroid as the **mean** of the points assigned to it.
5. **Repeat** 3–4 until the assignment stops changing (or centroids move less than a tolerance, or a max iteration count is hit).

Objective function

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist(x, c_i)^2$$

- K-means **minimises SSE** (within-cluster sum of squared error).
- The mean is provably the point minimising SSE for a cluster — **that is why step 4 uses the mean**.
- Complexity $O(nKI d)$: n points, K clusters, I iterations, d dimensions. Linear in n → scales well.

Choosing K 80 Ba 4

- **Elbow method**: plot SSE against K ; take the K where the curve bends. SSE always falls as K rises, so the **bend**, not the minimum, is the answer.
- **Silhouette coefficient**: pick the K maximising it.
- **Domain knowledge**; cross-validation; gap statistic.
- Rule of thumb $K \approx \sqrt{n/2}$.

Limitations (asked with almost every numerical)

- **K must be given in advance**.
- **Sensitive to the initial centroids** → different seeds give different clusterings; converges to a **local** optimum, not the global one. *Fix*: run several times and keep the lowest SSE, or use K-means++.
- **Sensitive to outliers**: the mean is dragged by extreme values. *Fix*: K-medoids.
- Only finds **convex, roughly spherical, similar-sized** clusters. Fails on elongated or nested shapes.
- Cannot handle **categorical** data (no mean). *Fix*: K-modes.
- Empty clusters can occur; needs a re-seeding rule.

K-medoids (PAM)

- Uses an **actual data point** (the medoid) as the cluster centre instead of the mean.
- Minimises the sum of *absolute* distances rather than squared distances.
- **Far more robust to outliers** and works with any distance measure.
- Cost: $O(K(n - K)^2)$ per iteration → much slower; CLARA samples to scale it.

Bisecting K-means 71 Ch 10

- Hybrid: produces a **hierarchy** using K-means as the engine.
- Steps:
 1. Start with all points in one cluster.
 2. Pick a cluster to split (largest, or highest SSE).
 3. Split it in two with K-means ($K = 2$); do this several times and keep the best split.
 4. Repeat until K clusters exist.
- **Less sensitive to initialisation** than plain K-means, and gives a dendrogram.

5.3 Hierarchical Clustering

What is hierarchical clustering? Agglomerative vs divisive. Draw the dendrogram. 2+7 8 4+4 10

HOT 5/23 80 Bh, 80 Ba, 79 Ba, 76 Ch, 70 Ch 4 of the 5 are numericals → companion.

Idea

- Produces a **nested tree of clusters**, not a single flat partition.
- **No need to specify K in advance** — cut the dendrogram at any height to get any K .
- **Dendrogram**: tree diagram where the **height of a merge** = **the distance between the two clusters merged**.

Two directions

- **Agglomerative (AGNES)** — **bottom-up**, the common one.
 1. Start: every point is its own cluster.
 2. Merge the **two closest** clusters.
 3. Update the distance matrix.
 4. Repeat until one cluster remains.
- **Divisive (DIANA)** — **top-down**.
 1. Start: all points in one cluster.
 2. Split the cluster that is least cohesive.
 3. Repeat until every point is alone (or a stopping rule fires).
- **Divisive is more expensive** ($2^{n-1} - 1$ ways to split), so agglomerative dominates.

Linkage: how to measure cluster-to-cluster distance (this is the whole exam question)

- **Single link (MIN)**: distance = **closest** pair between the 2 clusters.
 - + Finds **arbitrary, elongated** shapes.
 - - **Chaining effect**: a line of noise points can bridge 2 real clusters.
- **Complete link (MAX)**: distance = **farthest** pair.
 - + Less sensitive to noise, gives compact balanced clusters.
 - - Biased to globular shapes; **breaks large clusters**.
- **Average link**: mean of all pairwise distances. A compromise.
- **Centroid / Ward's**: distance between centroids / merge that increases SSE least.

Pros and cons

- + No K needed; the dendrogram is highly interpretable; any K obtainable afterwards.
- - $O(n^2)$ **space** and $O(n^3)$ **time** → does not scale.
- - **Merges are irreversible** — a bad merge can never be undone (greedy).

5.4 DBSCAN

What are core, border and noise points? Write the DBSCAN algorithm and how it handles noise. 3+5

8 8+2 7

TOP 12/23 82 Bh, 81 Bh, 79 Bh, 79 Ba, 78 Bh, 76 Ash, 75 Ch, 75 Ash, 74 Ash, 72 Ka, 70 Ch, 70 Asa

Two parameters

- ε (Eps) = the neighbourhood radius.
- **MinPts** = minimum number of points required inside that radius.
- ε -neighbourhood: $N_\varepsilon(p) = \{q \in D \mid \text{dist}(p, q) \leq \varepsilon\}$

Three kinds of point (learn these exactly)

- **Core point**: $|N_\varepsilon(p)| \geq \text{MinPts}$. Sits in the *interior* of a cluster.
- **Border point**: fewer than MinPts neighbours, **but lies within ε of a core point**. On the *edge*.
- **Noise point**: neither. **Discarded – this is how DBSCAN handles noise.**

The reachability definitions 74 Ash short note

- **Directly density-reachable**: q is directly density-reachable from p if p is a **core point** and $q \in N_\varepsilon(p)$.
- **Density-reachable**: there is a **chain** $p_1, \dots, p_n$ where each p_{i+1} is directly density-reachable from p_i . **Not symmetric** (the end may be a border point).
- **Density-connected**: p and q are density-connected if some object o exists from which **both** are density-reachable. **This one IS symmetric.**
- **A cluster** = a maximal set of density-connected points.

Algorithm

1. Label every point as core, border or noise by counting its ε -neighbours.
2. Discard the noise points.
3. Put an edge between every pair of core points within ε of each other.
4. Each **connected group of core points** becomes a cluster.
5. Assign each border point to a cluster of one of its associated core points.

Choosing the parameters 78 Bh asks this

- **k -distance plot**: for each point compute the distance to its k th nearest neighbour ($k = \text{MinPts}$), sort descending, plot. The **sharp bend (knee)** gives ε .
- Rule of thumb: $\text{MinPts} \geq d + 1$, commonly $2 \times d$ for d dimensions; $\text{MinPts} = 4$ for 2-D.

Pros

- Finds **arbitrarily shaped** clusters. • **K is not needed.**
- **Explicitly identifies noise / outliers** rather than forcing every point into a cluster.
- Insensitive to the order of points (except some border cases).

Cons and their fixes 76 Ash asks “how do we avoid the issues”

- **Fails on clusters of varying density** – one $(\varepsilon, \text{MinPts})$ pair cannot suit both a dense and a sparse cluster. **Fix**: **OPTICS**, which orders points across a range of ε .
- Very sensitive to the 2 parameters. **Fix**: the k -distance plot.
- Degrades in **high dimensions** (distance loses meaning). **Fix**: reduce dimensions first.
- A border point reachable from 2 clusters is assigned by processing order.

5.5 Issues: Evaluation, Scalability, Comparison

Explain the different measures of cluster validity. How do you evaluate the clusters generated? 10 4

3

HOT 6/23 80 Bh, 74 Ch, 72 Ka, 71 Ch, 70 Asa, 69 Ch

Why it is hard

- Clustering is unsupervised, so **there is no ground truth to compare against**.
- Any algorithm will return clusters even on **random data** — so the first question is whether cluster structure exists at all (**cluster tendency**).

Three families of measure

- **Internal (unsupervised)** — uses only the data.
 - **Cohesion** (compactness): within-cluster SSE. Lower is better.
 - **Separation**: between-cluster sum of squares. Higher is better.
 - **Silhouette coefficient** for point i :

$$s = \frac{b - a}{\max(a, b)}$$

a = mean distance to points in its own cluster, b = mean distance to the nearest other cluster. Range $[-1, 1]$; near **1** good, near **0** on a boundary, **negative** = **probably in the wrong cluster**.

- **External (supervised)** — compares against known labels.
 - **Purity, entropy, precision/recall/F, Rand index, Jaccard coefficient**.
- **Relative** — compares 2 clusterings, eg by SSE or silhouette, to choose K .

Other issues

- **Determining K** — see the elbow and silhouette methods in §5.2.
- **Scalability**: hierarchical is $O(n^3)$; K-means is linear but needs many passes; BIRCH and CLARA exist to scale.
- **High dimensionality**: distances converge → use subspace clustering (CLIQUE) or reduce dimensions first (PCA, Ch 2).
- **Mixed attribute types**: needs a mixed dissimilarity measure (Ch 2, §2.4).
- **Outliers** distort centroids → K-medoids or DBSCAN.
- **Interpretability**: a cluster is only useful if it can be described.

Choosing an algorithm 79 Bh 2+6

- Mixed attribute types (keywords, URLs, purchases, usage level) and unknown K → **use a method that needs no K and handles mixed data**: DBSCAN on a suitable dissimilarity measure, or hierarchical clustering.
- Validate with silhouette + domain inspection of the resulting segments.

5.6 PYQ Mapping

N = solved in the Ch 5 Numerical companion.

5.1 Basics

1. TOP 10/23 Cluster analysis + applications + types of clusters. 8 81 Ba 5 72 Ka 2+6 75 Ch 8 73 Shr 3+4 80 Bh
2. PIN 1/23 Why is clustering unsupervised? + hierarchical clusters via **Bisecting K-means**. 10 71 Ch
3. PIN 2/23 Types of clustering methods + DBSCAN w/ example 10 72 Ch; partition methods + hierarchical w/ example 10 74 Ash
4. PIN 1/23 What is a **contiguous cluster**? Explain an algorithm that generates them. 10 71 Shr
5. PIN 1/23 Internet-marketer segmentation: which algorithm, and how to validate. 2+6 79 Bh

5.2 K-means

6. PIN 1/23 K-means algorithm + strategy for choosing optimal K . 4+4 80 Ba
7. N Cluster for $K = 2$ w/ Euclidean distance + demerits. 5+3 81 Ba → **the paper prints NO data table; see the companion**.
8. N Two clusters from the 6-instance dataset. 2+6 78 Bh 2+8 74 Ash
9. N K-means w/ limitations, 6-row dataset. 4+6 75 Ash
10. N K-means w/ limitations, 7-row dataset. 10 71 Shr
11. N Matrix X , $K = 3$, given initial centres; final centres + iteration count. 8 76 Ch
12. PIN 1/23 Hierarchical vs partitioning + how K-means is applied, w/ example. 2+8 74 Ch

5.3 Hierarchical

13. **N** Draw the dendrogram for the 6×6 distance matrix. 2+7 80 Bh
14. **N** Complete-linkage clustering + dendrogram from coordinates. 8 80 Ba
15. **N** Single-link and complete-link dendrograms for 6 objects. 4+4 76 Ch
16. **PIN 1/23** Agglomerative vs divisive w/ illustrative examples. 10 70 Ch

5.4 DBSCAN

17. **TOP 4/23** Core / border / noise points + DBSCAN algorithm + noise handling. 3+5 79 Bh 8 75 Ch 8+2 76 Ash 7 72 Ka
18. **PIN 1/23** Neighbourhood quantification, how clusters are found, choosing Eps and MinPts, and the $\varepsilon_1 < \varepsilon_2$ subset proof. 2+2+2 78 Bh
19. **PIN 2/23** Density-based cluster + an algorithm that generates them. 8 70 Ch 8 75 Ash
20. **PIN 1/23** Short note: Density reachable and Density connected. 4 74 Ash

5.5 Issues

21. **PIN 1/23** Different measures of cluster validity. 10 71 Ch
22. **PIN 3/23** How do you evaluate the clusters generated? 4 80 Bh; short notes: Issues in clustering 74 Ch, Cluster evaluation 72 Ka. 3

Papers added by the 2026-09 merge

The six papers of `DM4.1BCT.pdf` (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. verified/added

- **N** Core, border and noise points + **DBSCAN on 8 points**, $\varepsilon = 2$, MinPts = 2. 3+5 82 Bh verified/added
- **N** K-means, 2 clusters from a 6-instance dataset. 2+8 82 Bh
- **N** K-means from **a single given centroid** (2, 3), $k = 2$, + problems and fixes. 5+3 81 Bh
- **N** K-means with the **Manhattan** distance, 3 centres, **stop after 3 iterations**. 8 79 Ba verified/added
- **N** Clustering vs classification + 1-D K-means on 7 points + **compute SSE**. 4+8 73 Ch verified/added
- Hierarchical clustering w/ a dendrogram example. 6 79 Ba
- What are density-based clusters? + DBSCAN w/ example. 10 70 Asa
- Measures that can be used to **compare two clusters** 70 Asa · issues in cluster validation 69 Ch. 6 6
- K-means w/ examples 69 Ch · short notes **DBSCAN clustering** 81 Bh, 79 Ba. 10 4 4

Coverage check

- All Ch 5 PYQs covered. **11 are numerical** → companion.
- **Known defect in the source** 81 Ba: Q4b says “cluster the following given data for $K = 2$ ” but **the paper prints no data table** — it jumps straight to Q5. Confirmed against the scan. The companion supplies a substitute dataset. verified/added
- **Gaps filled by research** verified/added: the **silhouette coefficient** formula, the **k -distance plot** for choosing ε , and the formal **density-reachable vs density-connected** distinction (your notes use the terms without defining them).

Sources used for this chapter

- **Han & Kamber 3rd ed** ch 10 (basic cluster analysis, partitioning, hierarchical, density-based, evaluation) and ch 11 (advanced). **Primary authority**.
- **DBK Lectures 15–16**, `data mining_/Chapter 5- Clustering.pdf`, `Clustering.pdf`, `ch7_ClusterAnalysis.ppt`, `Chapter7_3.ppt`, `kmedoids.doc`.

6. Anomaly / Fraud Detection

3 hrs · in 23/23 papers, the only chapter with perfect coverage · syllabus gives it no sub-topics · 1 numerical → Ch 6 companion.

Read this first:

- Only 3 hrs of syllabus, but it has appeared in **every single paper**. **Best marks-per-hour in the subject**.
- One question dominates: **“What is anomaly detection + explain distance-based methods”** **TOP 10/23**. Learn that and the importance/types answer, and the chapter is banked.

6.1 Basics

What is anomaly detection? Why is it important? 2+2+4 3+3 5 2+3

HOT 6/23 82 Bh, 81 Bh, 81 Ba, 79 Bh, 73 Ch, 72 Ka

Definition

- An **outlier / anomaly** is a data object that **deviates significantly** from the rest, as if generated by a **different mechanism**.
- **Anomaly detection** = finding those objects.
- Hawkins: an observation deviating so much that it arouses suspicion it was generated by a different process.

Outlier ≠ noise

- **Noise** = random error / variance in a measured value. Has **no meaning**, should be **removed** before mining.
- **Outlier** = a genuine object from a different mechanism. Carries **information**, and is often *the thing you are looking for*.
- **Same data point, opposite treatment** depending on the task.
- Related: **novelty detection** finds genuinely new patterns; once confirmed they are *folded into* the normal model, so later instances are no longer flagged.

Why it matters / applications

- **Fraud detection**: credit-card use unlike the customer's normal pattern.
- **Intrusion detection**: traffic unlike normal network behaviour.
- **Medical**: unusual symptoms or test results → disease.
- **Public health**: a spike in one region → outbreak.
- **Industry**: fault and defect detection, equipment failure.
- **Ecosystem disturbance**: hurricanes, floods, droughts.
- **Data cleaning**: find and remove genuinely erroneous values.

The one-line justification

- In these applications the **rare event is the valuable event**. Ordinary mining is tuned to find *frequent* patterns, so it actively discards exactly what matters here.

Types of outliers (80 Bh asks this directly, 5)

- **Global outlier (point anomaly)**
 - Deviates significantly from **the rest of the whole data set**.
 - Simplest type; most methods target these.
 - Eg a computer broadcasting a huge number of packets in a short time.
- **Contextual (conditional) outlier**
 - Deviates significantly **with respect to a chosen context**.
 - Attributes split into two groups:
 - o **Contextual attributes**: define the context. Eg date, location.
 - o **Behavioural attributes**: what is judged. Eg temperature, humidity.
 - Eg **28°C is an outlier in a Kathmandu winter, normal in summer**.
 - Generalises **local outliers**; a global outlier is the special case where the context is empty.
- **Collective outlier**
 - A **subset of objects that as a whole** deviates, even though **no individual member is an outlier**.
 - Eg 1 delayed shipment is normal; **100 delayed on one day** is a collective outlier.
 - Eg a burst of packets from many machines at once = denial-of-service, though each single packet looks fine.
 - Detecting them needs the **relationships** between objects (sequence, spatial, graph structure), not just the objects.

Challenges / issues 76 Ch 2, 72 Ch short note

- **Defining "normal"** is hard, and the boundary is rarely sharp.
- Normal behaviour **evolves** → today's model is stale tomorrow.
- Application-specific: one definition does not transfer between domains.
- **Scarcity of labelled data** for training and validation.
- **Noise looks like outliers** → hard to separate.
- **Interpretability**: you must justify *why* something was flagged.
- Handling **high dimensionality** (distances lose meaning).

6.2 Categories of Anomaly Detection Schemes

Briefly explain different types of anomaly detection schemes. 4 6

81 Ba, 78 Bh 78 Bh asks to compare and contrast three methods.

(A) By how much labelling is available

- **Supervised**
 - Training set labelled both normal *and* outlier → treat as a **classification** problem.
 - Problem: **severe class imbalance** (outliers are rare) + the sample may not represent the outlier distribution.
 - Fixes: oversample outliers, make up artificial outliers, cost-sensitive learning.
- **Semi-supervised**
 - Only the **normal** class is labelled (common in practice).
 - Build a model of normality; anything not matching it is an outlier.
- **Unsupervised**
 - **No labels at all**. Assumes normal objects are *far more frequent* and cluster together.
 - Most practical, and the default assumption in exam answers.

(B) By the assumption made (this is the "three methods" answer for 78 Bh)

Method	Assumption	Weakness
Statistical (model-based)	Normal data follows a stochastic model ; low-probability objects are outliers.	Must guess the right distribution; poor in high dims.
Proximity-based	An outlier is far from its neighbours (or in a sparse region).	$O(n^2)$ cost; sensitive to r and k ; struggles w/ varying density (distance-based only).
Clustering-based	Normal objects belong to large, dense clusters; outliers do not.	Depends on the clustering; costly; outliers can distort the clusters themselves.

- **Classification-based** is sometimes given as a 4th: train a 1-class model of normality (eg 1-class SVM), flag whatever falls outside.

6.3 Distance / Proximity-Based Approaches

Explain distance-based methods for anomaly detection. 2+3 8 2+6 10 5 4+3

TOP 10/23 82 Bh, 81 Bh, 80 Ba, 79 Ba, 75 Ch, 75 Ash, 74 Ch, 73 Shr, 71 Ch, 70 Asa The single most repeated question in this chapter.

Core idea

- Judge an object by its **proximity to its neighbours**.
- An object is an outlier if its neighbourhood is **sparse** or its neighbours are **far**.
- Needs only a distance measure → Ch 2, §2.4, no distribution assumption.

1. Distance-based: the $DB(r, \pi)$ outlier

- Fix a **distance threshold** r (defines the neighbourhood) and a **fraction threshold** π .
- Object o is a $DB(r, \pi)$ -outlier if:

$$\frac{|\{o' \mid \text{dist}(o, o') \leq r\}|}{\|D\|} \leq \pi$$

- In words: **fewer than a π fraction of all objects lie within distance r of o** .
- **Equivalent k -NN form**: with $k = \lceil \pi \|D\| \rceil$, object o is an outlier if $\text{dist}(o, o_k) > r$, where o_k is its k th nearest neighbour.
- **Nested-loop algorithm**: for each object, scan the others and count how many fall within r ; stop early once the count exceeds πn . Cost $O(n^2)$.

2. k -NN score (the other common phrasing)

- Score each object by the **distance to its k th nearest neighbour** (or the average of its k nearest).
- Rank all objects by that score; the top ones are the outliers.
- 71 Shr asks exactly this, 10.

Clustering-based 76 Ash short note

- Cluster the data, then flag as outliers the objects that:
 - belong to **no cluster** (eg DBSCAN noise points),
 - are **far from their own cluster centroid**, or
 - belong to a cluster that is **very small or very sparse**.
- **DBSCAN** gives it free: **noise points** are exactly the outliers. → Ch 5
- + Unsupervised; reuses clustering you may already have.
- – Effectiveness depends entirely on the clustering; outliers can themselves distort the clusters.

3. Density-based: the local outlier 80 Bh 3

- **Why needed**: distance-based methods take a **global** view, using one r for the whole data set.
- If the data has **one dense cluster and one sparse cluster**, no single r works:
 - Set r small → the entire sparse cluster is wrongly flagged.
 - Set r large → genuine outliers beside the dense cluster are missed.
- **Fix**: compare an object's density to **the density of its own neighbours**, not to the whole data set.
- **Local Outlier Factor (LOF)**:
 - $LOF \approx 1$ → density like its neighbours → normal.
 - $LOF \gg 1$ → much sparser than its neighbours → outlier.
 - Built from *reachability distance* and *local reachability density*.
- **Distance-based finds global outliers; density-based finds local ones**. That contrast is the marks.

Strengths / weaknesses

- + No distribution assumption; simple; works on any data w/ a distance measure.
- – $O(n^2)$ in the naive form.
- – Sensitive to the choice of r, π, k .
- – Degrades in high dimensions (curse of dimensionality).

Statistical approaches

- Assume a model, flag low-probability objects.
- **Parametric**: assume a distribution (usually **Gaussian**), estimate its parameters, flag points in the tails.
 - Common rule: more than **3 standard deviations** from the mean.
 - **Grubbs' test**: detects one outlier at a time in univariate normal data, removes it, repeats. H_0 : no outlier.
- **Non-parametric**: no assumed shape; use histograms or kernel density estimates.
- **Convex hull method** 70 Ch asks this by name verified/added
 - Points on the **convex hull** (the boundary of the smallest convex polygon enclosing all points) are assumed to be **extreme**, hence outliers.
 - Peel the hull layer by layer to get successive outlier candidates.
 - **Limitation**: only ever finds *boundary* points. An outlier sitting *inside* the data cloud, in a sparse pocket, is never detected.

The likelihood (mixture-model) approach 8

70 Asa “What is anomaly detection? Explain likelihood approach for anomaly detection.” Tan/Steinbach §10.2; not in Han & Kamber. verified/added

The model

- Treat the data set D as a **mixture of two distributions**:
 - M = the **majority** (normal) distribution, something tight such as a Gaussian.
 - A = the **anomaly** distribution, taken as **uniform** because nothing is assumed about how outliers are spread.
- λ = prior probability that an object is an anomaly, a small number fixed in advance.

$$D = (1 - \lambda) M + \lambda A$$

- At any moment the objects are partitioned into M_t and A_t . **Start with $M_0 = D$ and $A_0 = \emptyset$** : assume everything is normal.

Log-likelihood of the whole data set

$$\begin{aligned} LL_t(D) = & |M_t| \log(1 - \lambda) + \sum_{x_i \in M_t} \log P_{M_t}(x_i) \\ & + |A_t| \log \lambda + \sum_{x_i \in A_t} \log P_{A_t}(x_i) \end{aligned}$$

- First two terms are the *normal* account, last two the *anomaly* account.

Algorithm

1. $M \leftarrow D, A \leftarrow \emptyset$, compute $LL_t(D)$.
2. For each point $x \in M$:
 - (a) Tentatively move x to A .
 - (b) Re-estimate M 's parameters without x and compute the new $LL_{t+1}(D)$.
 - (c) $\Delta = LL_{t+1}(D) - LL_t(D)$.
 - (d) If $\Delta > c$ (a threshold), **keep x in A and declare it an anomaly**; otherwise put it back in M .
3. Repeat until no point moves.

Why the test works (this is the explanation the marks are for)

- Moving a *normal* point out forces M to explain fewer points it already explained well, and the uniform A explains it badly $\rightarrow$ likelihood **falls**.
- Moving a *genuine outlier* out lets M tighten around the rest, and the outlier was costing M a huge penalty anyway $\rightarrow$ likelihood **rises sharply**.
- So **a large gain in likelihood is itself the evidence** that the point does not belong.

Strengths and weaknesses

- + Firm statistical footing; gives a principled test rather than an arbitrary distance.
- + Handles mixed / categorical attributes if a model exists for them.
- – Needs a distribution assumption for M , and a wrong one gives wrong outliers.
- – Expensive: M is re-estimated once per candidate point.
- – The threshold c and the prior λ are both guesses.
- – Degrades badly in **high dimensions**, like every parametric method.

6.4 Base Rate Fallacy

Base rate fallacy 3 5

70 Ch short note, 69 Ch 7, 76 Ch numerical **Numerical solved in the Ch 6 companion.** Not in Han & Kamber; from Tan/Steinbach and Axelsson.
verified/added

The fallacy

- **Base rate** = the prior probability of the rare event, eg $P(\text{intrusion})$.
- **Base rate fallacy** = the tendency to **ignore the base rate** and judge only by the test's accuracy.
- Consequence: when the event is very rare, **even a highly accurate detector produces mostly false alarms**, because the huge normal population generates more false positives than the tiny anomalous population generates true positives.

Set-up (intrusion detection)

- I = intrusive behaviour, $\neg I$ = normal. A = alarm, $\neg A$ = no alarm.
- **Detection rate** = $P(A|I)$ (true positive rate)
- **False alarm rate** = $P(A|\neg I)$
- **Bayesian detection rate** = $P(I|A) \rightarrow$ **what actually matters to the analyst: given an alarm, is it real?**

Bayes

$$P(I|A) = \frac{P(A|I)P(I)}{P(A|I)P(I) + P(A|\neg I)P(\neg I)}$$

Worked illustration

- Test 99% accurate both ways, disease prevalence 1 in 100.
- Then $P(\text{disease} | \text{positive})$ is only about **1/2**, not 99%.
- With a rarer event the result gets far worse.

Axelsson's conclusion

- In intrusion detection $P(I)$ is **tiny**, so the false alarm rate **dominates** the equation.
- $\rightarrow$ you need an **extremely low false alarm rate** to get a usable Bayesian detection rate. Raw accuracy is not enough.
- This is the formal reason **accuracy is the wrong metric** for anomaly detection $\rightarrow$ same argument as class imbalance in Ch 3, §3.7.

6.5 PYQ Mapping

N = solved in the Ch 6 Numerical companion. The syllabus gives this chapter no sub-topics, so PYQs are grouped by what they ask.

1. TOP 6/23 Anomaly detection + distance-based methods. The core question.

- "What is anomaly detection? Explain distance based method." 2+3 80 Ba
- same wording 8 73 Shr
- "Why is it important? + distance based method" 2+6 75 Ash
- "What do you mean by anomaly detection and why is it important? Describe distance based approaches." 4+3 74 Ch
- "Explain few distance based approaches" 10 71 Ch
- "What is outlier? Explain the distance base approaches." 5 75 Ch

2. HOT 3/23 Definition + importance + types of schemes.

- "What is it? Why important? Briefly explain different types of schemes." 2+2+4 81 Ba
- "Why is it important and where is it applicable?" 3+3 79 Bh
- "Discuss its importance in security." 5 72 Ka

3. PIN 1/23 Types of outlier w/ examples + how density-based detection works.

5+3 80 Bh

4. PIN 1/23 Compare and contrast 3 different methods of anomaly detection.

6 78 Bh

5. PIN 1/23 Issues related to anomaly detection 2 76 Ch; N false alarm rate / detection rate numerical

3 76 Ch

6. PIN 1/23 How can the Nearest-Neighbor algorithm be used for anomaly detection?

10 71 Shr

7. PIN 4/23 Short notes:

- Clustering and its application in anomaly detection 3 76 Ash
- Issues in anomaly / fraud detection 4 72 Ch
- Data Mining for Anomaly Detection 4 74 Ash
- **Anomaly Detection; Base Rate Fallacy; Convex Hull Method** 15 70 Ch

Papers added by the 2026-09 merge

The six papers of [DM4.1BCT.pdf](#) (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. verified/added

- What is anomaly detection? + distance-based method. 2+5 82 Bh
- Anomaly detection + the different **approaches**. 2+6 81 Bh
- **Strengths and weaknesses** of the statistical, proximity-based, density-based and cluster-based approaches. 6 79 Ba
- Anomaly detection + the **issues** associated with it. 2+3 73 Ch
- Anomaly detection + the **likelihood approach**. 8 70 Asa verified/added
- What is **Base Rate Fallacy**? Explain w/ example. 7 69 Ch

Coverage check

- All Ch 6 PYQs covered. 1 numerical 76 Ch → companion.
- **Gaps filled by research** verified/added :
 - **Convex hull method** 70 Ch: named in the Tan slides but never explained, and absent from Han & Kamber. Written out above w/ its limitation.
 - **Base rate fallacy**: the notes give the psychology definition but not the intrusion-detection form. Axelsson's argument added.
 - **Grubbs' test** and the supervised/semi-supervised/unsupervised split.

Sources used for this chapter

- **Han & Kamber 3rd ed**, ch 12 Outlier Detection: definition, the 3 outlier types, method taxonomy, $DB(r, \pi)$ formula, density-based / LOF. **Primary authority**.
- [Anomaly Detection/Anomaly Detection.pdf](#) (Tan, Steinbach & Kumar slides): convex hull, Grubbs' test, base rate fallacy, Axelsson.
- [data mining_/Chapter 6 _ 7.pdf](#) (Bhupesh K. Mishra), [Intrusion.pptx](#), **DBK Lecture-17**.

7. Advanced Applications

3 hrs · in 22/23 papers · 3 topics · 1 numerical (PageRank) → Ch 7 companion.

Weight of the chapter:

- Web mining **TOP 11/23** → **3 hrs of syllabus, asked in 11 papers. Best marks-per-hour after Ch 6.**
- Time-series data mining **HOT 6/23** · Mining object & multimedia **PIN 3/23**
- Mostly **short notes**, so breadth beats depth here.

7.1 Web Mining

What is Web Mining? Briefly explain the structure / taxonomy of Web Mining. **3+5 6 8**

TOP 14/23 81 Ba, 80 Bh, 79 Bh, 79 Ba, 78 Bh, 76 Ch, 76 Ash, 75 Ch, 74 Ch, 74 Ash, 73 Ch, 73 Shr, 70 Ch, 69 Ch

Definition

- **Web mining** = applying data mining to discover patterns from **web data**: the content, the link structure and the usage logs.

Why the Web is hard (good opening lines)

- Huge, growing, and **widely distributed**.
- **Semi-structured / unstructured** (HTML, not tables).
- **Highly dynamic**; heterogeneous; multilingual.
- Only a **tiny fraction** is relevant to any one user.
- Broad topics return hundreds of thousands of documents → **search returns too much, and misses too much.**

The taxonomy (3 branches — this IS the answer)

- 1. Web Content Mining:** mine the *content* of pages (text, images, video).
 - *Web page content mining*: extract facts from within pages.
 - *Search result mining*: summarise / cluster / categorise results.
- 2. Web Structure Mining:** mine the *hyperlink graph*.
 - Links carry authority: a link is a **vote**.
 - Algorithms: **PageRank**, **HITS**, **CLEVER**.
 - Finds authoritative pages, hubs, and web communities.
- 3. Web Usage Mining:** mine the *server logs* / click-streams.
 - *General access pattern tracking*: how the site is used overall.
 - *Customised usage tracking*: per-user personalisation.
 - Uses: site restructuring, recommendation, targeted ads, caching, fraud detection.

PageRank (asked as a short note in 2 papers + 1 numerical)

- Ranks a page by **who links to it**, not by what it says.
- **Two ideas**:
 1. A link from page T to page A is a **vote** for A .
 2. Votes are **weighted by the voter's own rank**, and **split** across all its outgoing links.
- **Recursive**: a page is important if important pages point to it.

$$PR(A) = (1 - d) + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

- T_i = pages linking to A ; $C(T_i)$ = number of **outbound** links from T_i .
- d = **damping factor**, usually **0.85**.
- **Random surfer model**: with probability d the surfer follows a link, with probability $(1 - d)$ they jump to a random page.
- **Why damping is needed**: without it, a group of pages linking only to each other would hoard all the rank (a "rank sink"), and pages with no outlinks would leak it.
- Computed **iteratively** until the values settle; ranks converge quickly.

Text mining 81 Ba 4

- **Key steps**:
 1. **Text preprocessing**: tokenise, remove **stop words**, **stemming** (reduce to root form), case folding.
 2. **Feature/attribute generation**: build the **term vector** (bag of words).
 3. **Feature selection**: drop uninformative terms; weight by **TF-IDF**.
 4. **Mining**: classification, clustering, association, topic modelling.
 5. **Analysis / interpretation** of the results.
- Similarity between documents → **cosine similarity** → Ch 2, §2.4.
- **Information retrieval** measures: precision and recall → Ch 3, §3.7.

Challenges of web mining 75 Ch 5

- Scale and growth rate; dynamic content; the deep/hidden web.
- Noise: ads, navigation bars, boilerplate.
- **Spam**: link farms deliberately gaming the ranking.
- Privacy concerns in usage mining; no schema; duplicates and mirrors.

HITS (for contrast)

- Two scores per page: **authority** (pointed to by good hubs) and **hub** (points to good authorities).
- **Query-dependent** and computed at query time; PageRank is query-independent and precomputed.

Mining object and multimedia data 6 3 4

PIN 3/23 75 Ash, 74 Ch, 75 Ch

- **Multimedia data** = image, audio, video, text-plus-media in digital form.
- **Feature extraction first** — you never mine the raw pixels:
 - Image: colour histogram, texture, shape, edges.
 - Audio: pitch, frequency, MFCC. Video: motion vectors, key frames.
- **Tasks**: similarity search (content-based retrieval), classification, clustering, association among media features, video-shot detection.
- **Challenges**: very high dimensionality, huge volume, and the **semantic gap** — low-level features do not map cleanly to human meaning.

Visual data mining 75 Ch short note 4

- Uses **visualisation** to help the user discover knowledge, combining human perception with machine computation.
- **Categories**: data visualisation, mining-*result* visualisation, mining-*process* visualisation, and interactive visual mining.
- Techniques: scatter plots, box plots, parallel coordinates, pixel-oriented, dendrograms.
- **Why it works**: humans spot patterns and outliers in a picture far faster than in a table.

7.2 Time-Series Data Mining

Explain time series data mining with an example. 6 5 4 3

TOP 11/23 82 Bh, 81 Bh, 80 Ba, 79 Ba, 75 Ch, 74 Ch, 73 Ch, 72 Ch, 72 Ka, 71 Shr, 70 Asa Mostly short notes, but **now asked in 11 papers**.

Definition

- A **time series** is a sequence of values recorded at **successive, usually equal, time intervals**.
- **Time-series data mining** = finding patterns, trends and regularities in such data.
- Eg stock prices, daily temperature, sales per month, sensor readings, ECG, network traffic, power consumption.
- **What makes it different:** the **order matters** and consecutive values are **correlated**, so the usual “independent tuples” assumption breaks.

The 4 components of a time series (memorise: T-C-S-I)

- **Trend (T):** the long-term overall direction. Eg steady sales growth.
- **Cyclic (C):** long, irregular swings, over years. Eg business cycles.
- **Seasonal (S):** regular, calendar-fixed variation. Eg umbrella sales in monsoon.
- **Irregular / random (I):** residual noise left after the other three.
- Models: multiplicative $Y = T \times C \times S \times I$, or additive $Y = T + C + S + I$.

Tasks

- **Trend analysis + forecasting** (moving average, exponential smoothing, ARIMA).
- **Similarity search:** find series that behave alike.
 - *Whole-sequence* matching vs *subsequence* matching.
 - Needs **normalisation** (offset and amplitude) first, else scale dominates.
 - **Dynamic Time Warping (DTW)** aligns series that are out of phase or run at different speeds — Euclidean distance cannot. verified/added
- **Motif discovery:** repeated subsequences. **Anomaly detection:** unusual subsequences → Ch 6.
- **Segmentation and classification** of series.

Smoothing: moving average

- Replace each value by the mean of a window around it → removes the irregular component and exposes the trend.
- A larger window smooths more but lags more.

Example to quote

- Daily closing prices of a share over 5 years: the **trend** is the multi-year rise, the **seasonal** part is the recurring festival-quarter bump, the **cyclic** part is the multi-year market swing, and a one-day crash is **irregular**. Mining finds the trend for forecasting and the crash as an anomaly.

7.3 PYQ Mapping

N = solved in the Ch 7 Numerical companion.

7.1 Web mining

1. **TOP 4/23** What is web mining + its structure / categories / taxonomy. 3+5 80 Bh 6 74 Ash 8 76 Ash 5 75 Ch
2. **PIN 1/23** Key steps in **text mining** + how to find page rank. 4+4 81 Ba
3. **N PageRank** with damping factor 0.85, 5 iterations, initial rank 0.25. 8 76 Ch
4. **PIN 2/23** Short notes: Page Rank algorithm 5 79 Bh 4 78 Bh
5. **HOT 3/23** Short note: Web mining 4 73 Shr 3 74 Ch 70 Ch

7.2 Mining object and multimedia

6. **PIN 1/23** Explain Web mining and Multimedia mining. 6 75 Ash
7. **PIN 2/23** Short notes: Multimedia mining 3 74 Ch; Visual Data Mining 4 75 Ch

7.3 Time-series

8. **PIN 1/23** Explain time series data mining in brief. 6 72 Ka
9. **PIN 1/23** Challenges of web mining + time series data mining w/ example. 5 75 Ch
10. **HOT 4/23** Short note: Time Series Data Mining. 5 80 Ba 3 74 Ch 4 72 Ch 71 Shr

Papers added by the 2026-09 merge

The six papers of `DM4.1BCT.pdf` (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. verified/added

- Explain **seasonality** in time-series data. 5 70 Asa
- Short notes: **Time Series Data Mining**. 5 4 4 4 82 Bh, 81 Bh, 79 Ba, 73 Ch
- Short notes: **Web mining** 79 Ba · **www mining** 73 Ch · **Page Rank** 69 Ch. 4 4 5
- Short note: **Data visualization**. 4 79 Ba

Coverage check

- All Ch 7 PYQs covered. 1 numerical 76 Ch → companion.
- **This chapter is 3 hrs of syllabus but appears in 16/17 papers**, almost entirely as **4–6 mark short notes**. Learn the **web mining taxonomy**, the **PageRank formula** and the **4 time-series components** and you can answer nearly every one of them.
- **Gaps filled by research** verified/added : the **PageRank formula** and the **random-surfer justification for damping** (your slides name PageRank but never give the equation); **DTW**; the **T-C-S-I decomposition**; the **semantic gap** in multimedia mining.

Sources used for this chapter

- **Han & Kamber 3rd ed** ch 13 (trends and applications) and the mining-complex-data material. **Primary authority**.
- `Chapter9_0.ppt` (web mining taxonomy), `Chapter9_1.ppt` (multimedia), `Chapter9_2.ppt` (text and web), `a1VisualMine.ppt` (visual data mining), `data mining_/Chapter 6 _ 7.pdf`, **DBK Lectures 18–19**.

8. Last-Minute Revision

Built from the frequency analysis of all 17 papers (2070–2081 BS). Nothing here is new material: everything points back into the chapters.

R.1 Must Know (the 15 TOP-tier topics — these carry the paper)

Ch	Topic	Papers	The one thing to be able to do
6	Anomaly detection + distance-based methods	17/17	Define outlier vs noise; $DB(r, \pi)$; distance vs density-based
2	Data pre-processing, 5 major tasks	12/17	Name the 5 tasks, 1 technique each
4	FP-Growth / FP-Tree	11/17	Build the tree, then mine it bottom-up
7	Web mining	11/17	The 3-branch taxonomy + PageRank formula
2	OLAP & multidimensional analysis	10/17	The 5 OLAP operations + OLAP vs OLTP
5	Cluster analysis basics & types	10/17	Types of clusters; clustering vs classification
1	DW fundamentals	9/17	S-I-T-N + three-tier architecture
3	Decision tree classifier	9/17	Information gain, full tree build
4	Frequent itemsets & Apriori	9/17	The Apriori property + join/prune/count
3	Bayesian classifier	8/17	Naive Bayes numerical + Laplacian correction
5	K-means	8/17	Run it to convergence, state limitations
5	DBSCAN	8/17	Core/border/noise + density-reachable vs connected
1	DM origin / KDD process	7/17	The 7 KDD steps + the pipeline diagram
3	Overfitting, validation, model comparison	7/17	Confusion matrix, all 6 metrics
4	Association basics	7/17	Support, confidence, market basket

R.2 Must Memorize (formulas you cannot derive in the hall)

Ch 2 – preprocessing & similarity

- Min-max: $v' = \frac{v - \min_A}{\max_A - \min_A} (n_{\max} - n_{\min}) + n_{\min}$
- Z-score: $v' = \frac{v - \bar{A}}{\sigma_A}$ **mean, not min**
- Decimal scaling: $v' = v/10^j$, $\max |v'| < 1$
- Equal-width bin: $W = (B - A)/N$ **max minus min**
- Minkowski: $(\sum |p_k - q_k|^r)^{1/r}$; $r=1$ Manhattan, $r=2$ Euclidean, $r \rightarrow \infty$ supremum
- $SMC = \frac{q+t}{q+r+s+t}$, $J = \frac{q}{q+r+s}$ **Jaccard drops t**
- $\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}$
- Nominal: $d = (p - m)/p$ Ordinal: $z = \frac{r-1}{M-1}$
- PCA variance explained = $\lambda_i / \sum \lambda_j$; $\sum \lambda = \text{trace}$
- n dims $\rightarrow 2^n$ cuboids

Ch 3 – classification

- $Info(D) = -\sum p_i \log_2 p_i$
- $Info_A(D) = \sum \frac{|D_j|}{|D|} Info(D_j)$;
 $Gain = Info(D) - Info_A(D)$
- $SplitInfo_A = -\sum \frac{|D_j|}{|D|} \log_2 \frac{|D_j|}{|D|}$;
 $GainRatio = \frac{Gain}{SplitInfo}$
- $Gini(D) = 1 - \sum p_i^2$ **binary splits only**
- Bayes: $P(H|X) = \frac{P(X|H)P(H)}{P(X)}$; naive:
 $P(X|C_i) = \prod P(x_k|C_i)$
- Laplace correction: $\frac{n_k+1}{|C_i,D|+k}$
- Sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$, $\sigma' = \sigma(1 - \sigma)$
- Rule: $Laplace = \frac{n_{correct}+1}{n_{covers}+k}$

Ch 3 – the metric table (8 papers)

- Accuracy = $\frac{TP+TN}{P+N}$ Error = 1–accuracy
- Sensitivity = recall = $TPR = \frac{TP}{P}$
- Specificity = $TNR = \frac{TN}{N}$
- FPR = false alarm = $\frac{FP}{N} = 1 - \text{specificity}$
- Precision = $\frac{TP}{TP+FP}$
- $F_1 = \frac{2 \text{prec} \cdot \text{rec}}{\text{prec} + \text{rec}}$
- $P = TP + FN$ (actual positives), $N = FP + TN$

Ch 4 – association

- $\text{support}(A \Rightarrow B) = P(A \cup B)$
- $\text{confidence}(A \Rightarrow B) = \frac{\text{sup}(A \cup B)}{\text{sup}(A)}$
- $\text{lift} = \frac{P(A \cup B)}{P(A)P(B)} > 1$ positive, < 1 negative
- $\chi^2 = \sum \frac{(\text{obs} - \text{exp})^2}{\text{exp}}$
- $\text{all_conf} = \frac{\text{sup}(A \cup B)}{\max\{\text{sup}(A), \text{sup}(B)\}}$;
 $\text{Kulc} = \frac{1}{2}(P(A|B) + P(B|A))$
- k -itemset $\rightarrow 2^k - 2$ candidate rules

Ch 5 – clustering

- $SSE = \sum_i \sum_{x \in C_i} \text{dist}(x, c_i)^2$
- Silhouette $s = \frac{b-a}{\max(a,b)}$
- DBSCAN: core if $|N_\epsilon(p)| \geq \text{MinPts}$

Ch 6 & 7

- DB(r, π)-outlier: $\frac{\|\{o' | \text{dist}(o, o') \leq r\}\|}{\|D\|} \leq \pi$
- $P(I|A) = \frac{P(A|I)P(I)}{P(A|I)P(I) + P(A|\neg I)P(\neg I)}$
- $PR(A) = (1 - d) + d \sum \frac{PR(T_i)}{C(T_i)}$, $d = 0.85$

R.3 Must Practice (numericals, ranked by how often they appear)

Ch	Numerical	Papers	Trap that costs the marks
3	Confusion matrix metrics	8	Papers flip actual/predicted between years. Label the axes first
5	K-means	7	Seed choice changes the answer; state your seeds
4	Apriori	7	Convert % to a count first; show the prune step
4	FP-tree + FP-Growth	6	Reorder each transaction into F-list order; state the tie-break
5	Hierarchical dendrogram	4	Single = MIN, complete = MAX; heights never decrease
3	Decision tree (ID3)	3	A pure partition contributes $Info = 0$
2	Similarity / distance	3	Normalise before cosine on mixed scales
3	Naive Bayes	2	Any zero $\rightarrow$ Laplacian correction
2	Normalisation	1	j must make $\max v' $ strictly < 1
2	PCA	1	Eigenvalues must sum to the trace
3	KNN	1	Show the whole distance table
7	PageRank	1	Split a page's rank across <i>all</i> its outlinks
6	Base rate fallacy	1	The prior is the point; compute $P(I A)$
3	ANN gradients	1	Chain rule; x_2 and u_1 must both appear

R.4 Most Repeated PYQs (if you only revise 12 questions, revise these)

1. What is anomaly detection? Explain distance-based methods. 2+3 8 10 6 papers
2. What is FP-Growth? Explain with example. 2+5 8 10 7 papers
3. What is data mining? Explain the steps of the KDD process. 2+3 2+6 7 papers
4. Explain typical OLAP operations + differentiate OLAP and OLTP. 6+4 10 4 papers
5. Confusion matrix → accuracy, sensitivity, specificity, precision. 4 10 8 papers
6. Core, border and noise points + the DBSCAN algorithm. 3+5 8 4 papers
7. What is cluster analysis? Applications + types of clusters. 8 5 5 papers
8. Decision tree + how information gain selects attributes. 8 10 9 papers
9. Explain Naive Bayesian classification with a suitable example. 6 8 10 8 papers
10. What is web mining? Explain its taxonomy. 3+5 6 8 4 papers
11. Apriori algorithm + generate frequent itemsets from the given table. 8 10 7 papers
12. Why data preprocessing? Explain its major tasks. 2+7 4+4 3 papers

Marks arithmetic. Chapters 3, 4 and 5 (classification, association, clustering) together carry **31 of the 45 syllabus hours** and supply most of the numericals. Chapters 1, 6 and 7 are only 8 hours combined but appear in **16–17 of 17 papers** as short, high-yield theory. **Do not skip the small chapters** — they are the cheapest marks in the paper.

R.5 One-Day Revision Checklist

Morning — theory (about 4 hrs)

- ☐ **Ch 1** (20 min): KDD 7 steps + diagram · S-I-T-N · three-tier DW architecture · 4 coupling schemes
- ☐ **Ch 2 theory** (40 min): 5 preprocessing tasks · missing-value methods · 5 OLAP operations · OLAP vs OLTP table · attribute types
- ☐ **Ch 3 theory** (60 min): 2-step classification · Hunt's algorithm · the 3 selection measures · pruning · rule-based + conflict resolution · Naive Bayes vs BBN · ANN backprop steps · validation methods + ROC
- ☐ **Ch 4 theory** (40 min): support/confidence · Apriori property · Apriori vs FP-Growth table · lift · categorical issues · sequential + subgraph
- ☐ **Ch 5 theory** (40 min): types of clusters · K-means limitations · single vs complete link · DBSCAN definitions · cluster validity
- ☐ **Ch 6** (30 min): **highest yield per minute** — outlier vs noise · 3 outlier types · 3 method families · $DB(r, \pi)$ · convex hull · base rate fallacy
- ☐ **Ch 7** (25 min): web mining taxonomy · PageRank · text mining steps · T-C-S-I · multimedia + visual mining

Afternoon — numericals (about 4 hrs)

- ☐ **Confusion matrix** (30 min): do all 8 from the Ch 3 companion. **Do the 75 Ash / 72 Ch pair back to back** to burn in the orientation trap
- ☐ **Apriori** (45 min): 80 Bh, 80 Ba, 76 Ch
- ☐ **FP-Growth** (45 min): 81 Ba end to end, then 78 Bh tree
- ☐ **K-means** (45 min): 78 Bh both seedings, then 76 Ch with $K = 3$
- ☐ **Hierarchical** (30 min): 76 Ch single *and* complete link
- ☐ **Decision tree** (30 min): 79 Bh full build
- ☐ **Naive Bayes** (20 min): 72 Ch, then 79 Bh with the Laplacian correction
- ☐ **Quick set** (25 min): normalisation · PCA · similarity · PageRank · base rate

Evening — consolidation (about 1 hr)

- ☐ Re-read **R.2 Must Memorize** twice. Write the formulas from memory once.
- ☐ Re-read every **orange** line in the chapters — they are the traps.
- ☐ Skim the **PYQ mapping** at the end of each chapter to see the wording variants examiners use.
- ☐ Check the **abbreviation chart** so nothing in your own notes is unfamiliar.

In the hall

- ☐ Confusion matrix → **label the axes before computing**
- ☐ Association → **convert % to a count first**
- ☐ K-means → **state your initial centroids**
- ☐ Every numerical → **write the formula before substituting**
- ☐ Diagrams earn marks: draw the KDD pipeline, the FP-tree, the dendrogram

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.